

3

Canonical and grand canonical ensembles: applications

This chapter is devoted to some important applications of the canonical and grand canonical formalisms. We shall concentrate on situations where the effects of quantum statistics, to be studied in Chapter 5, may be considered as negligible. The use of the formalism is quite straightforward when one may neglect the interactions between the elementary constituents of the system. Two very important examples are the ideal gas (non-interacting molecules) and paramagnetism (non-interacting spins), which are the main subjects of Section 3.1. In the following section, we show that at high temperatures one may often use a semi-classical approximation, which leads to simplifications in the formalism. Important examples are the derivation of the Maxwell velocity distribution for an interacting classical gas and of the equipartition theorem. These results are used in Section 3.3 to discuss the behaviour of the specific heat of diatomic molecules. In Section 3.4 we address the case where interactions between particles cannot be neglected, for example in a liquid, and we introduce the concept of pair correlation function. We show how pressure and energy may be related to this function, and we describe briefly how it can be measured experimentally. Section 3.5 shows the fundamental rôle played by the chemical potential in the description of phase transitions and of chemical reactions. Finally Section 3.6 is devoted to a detailed exposition of the grand canonical formalism, including a discussion of fluctuations and of the equivalence with the canonical ensemble.

3.1 Simple examples in the canonical ensemble

3.1.1 Mean values and fluctuations

In the preceding chapter, we defined the canonical ensemble as the equilibrium distribution (2.64) with only one relevant operator, the Hamiltonian, and consequently only one Lagrange multiplier $\lambda_1 = -\beta$ ($\beta = 1/kT$). Recall the main results of

Sections 2.4.4 and 2.5.4. The density operator D_B and the partition function are given by

$$D_B = \frac{1}{Z} e^{-\beta H} = \frac{1}{Z} \sum_r |r\rangle e^{-\beta E_r} \langle r| \quad (3.1a)$$

$$Z = \text{Tr} e^{-\beta H} = \sum_r e^{-\beta E_r} \quad (3.1b)$$

$$Z = \int dE \rho(E) e^{-\beta E} \quad (3.1c)$$

with $\beta = (kT)^{-1}$, $H|r\rangle = E_r|r\rangle$, $\rho(E)$ is the level density and we draw the reader's attention to the very important Footnote 15 in Chapter 2.

In the canonical ensemble, we know with precision the values of a set of external parameters¹ $x = \{x_i\}$, controllable from outside the system under consideration, for example the volume, the electric or magnetic field components, etc. We now undertake to explain precisely the rôle they play. The Hamiltonian depends on these external parameters: $H = H(x)$, but only its average value, $E(\beta, x)$ is known. According to (2.66), and writing explicitly the external parameters, $E(\beta, x)$ can be written as

$$E(\beta, x) = - \left. \frac{\partial \ln Z}{\partial \beta} \right|_x \quad (3.2)$$

For a quasi-static process, and for each external parameter x_i , we can obtain, with the help of the equilibrium distribution (3.1) and Equation (2.95), the average value of the corresponding conjugate variable (or 'conjugate force') X_i

$$X_i = \left\langle \frac{\partial H}{\partial x_i} \right\rangle = - \frac{1}{\beta} \left. \frac{\partial \ln Z}{\partial x_i} \right|_{\beta, x_{j \neq i}} \quad (3.3)$$

We remark that in calculating macroscopic quantities at thermodynamic equilibrium, $\ln Z$ plays the central rôle, and not Z . The logarithm of the partition function is in fact the 'good' thermodynamic potential for a system described by the canonical distribution: $\ln Z$ is an extensive quantity, which is not true for Z . Equating, in this chapter, the Boltzmann entropy with the thermodynamic entropy (cf. Section 2.5.2) we can write Equation (2.67) as

$$\ln Z = \frac{S}{k} - \beta E \quad (3.4)$$

¹ We reserve the term 'external parameter' for quantities that appear in Equation (2.94) for the work. The number of particles, N , which might appear at first sight to be an external parameter in the canonical ensemble, does not satisfy our definition.

This expression allows us to complete (2.68b) by adding the contributions of the conjugate forces. Using Equations (3.2) and (3.3) we obtain the entropy differential

$$\begin{aligned}\frac{dS}{k} &= d \ln Z + d(\beta E) = -E d\beta - \beta \sum_i X_i dx_i + \beta dE + E d\beta \\ &= \beta dE - \beta \sum_i X_i dx_i\end{aligned}$$

which establishes the general form of the ‘ TdS equation’ in the canonical ensemble²

$$TdS = dE - \sum_i X_i dx_i \quad (3.5)$$

When we use the variable T instead of β , the relevant thermodynamic potential is represented by the free energy $F(T, x)$

$$F = -\frac{1}{\beta} \ln Z \quad dF = -S dT + \sum_i X_i dx_i \quad (3.6)$$

The conjugate forces are obtained by taking the derivatives of the free energy with respect to external parameters. As an example, we obtain the pressure by considering $-V$ as an external parameter

$$P = \frac{1}{\beta} \left. \frac{\partial \ln Z}{\partial V} \right|_{\beta, N} = - \left. \frac{\partial F}{\partial V} \right|_{T, N} \quad (3.7)$$

Although N is not considered an external parameter, we can clearly calculate the chemical potential in the canonical ensemble by using its thermodynamic definition (1.11) and Equation (3.6)

$$\mu = \left. \frac{\partial F}{\partial N} \right|_{T, V} = - \left. \frac{1}{\beta} \frac{\partial \ln Z}{\partial N} \right|_{\beta, V} \quad (3.8)$$

To conclude, we examine fluctuations in the canonical ensemble. The fluctuation-response theorem can be written here as

$$\langle H^2 \rangle - E^2 = \langle (H - E)^2 \rangle = \left. \frac{\partial^2 \ln Z}{\partial \beta^2} \right|_x \quad (3.9)$$

² This equation can be also obtained from (3.4), which shows that $\ln Z$ is the Legendre transform of the entropy with respect to the energy: $\ln Z$ is a Massieu function.

$$\ln Z \equiv \Phi'_1(\beta, x) = \frac{1}{k} \Phi_1(1/T, x) \quad d \ln Z = -E d\beta - \beta \sum_i X_i dx_i$$

where Φ_1 is the Massieu function defined in (1.29); also see Exercise 1.6.1.

By using (3.2) and also taking $\Delta H = H - E$, Equation (3.9) becomes

$$\langle (\Delta H)^2 \rangle = - \left. \frac{\partial E}{\partial \beta} \right|_x = kT^2 C_x \geq 0 \quad (3.10)$$

where C_x is the specific heat at constant x (generally C_V). This demonstrates the positivity of C which is required for stability (cf. (1.58)).

We see in (3.2) and (3.9) the rôle played by $\ln Z$ in generating, by differentiation, the moments of physical quantities. However, we caution against a hasty generalization³

$$\langle (\Delta H)^n \rangle \neq \left. \frac{\partial^n \ln Z}{\partial \beta^n} \right|_x \quad \text{for } n > 3$$

3.1.2 Partition function and thermodynamics of an ideal gas

The canonical ensemble provides us the occasion to re-examine the concept of an ideal gas, introduced in Chapter 2, and to discuss its fundamental hypotheses. As we saw in Section 2.1.3, the quantum state of a molecule in a mono-atomic gas is determined by three integer quantum numbers (n_x, n_y, n_z) that give the components of the wave vector $\vec{k}$: $k_\alpha = (2\pi n_\alpha)/L_\alpha$. In the ideal gas approximation, the energy (entirely kinetic) of a molecule is given by $\varepsilon \sim E/N$. The *average occupation number*⁴ $\langle n_r \rangle$ of a level r , i.e. the average number of molecules occupying the energy level ε_r , is of the order of $N/\Phi(\varepsilon)$, where $\Phi(\varepsilon)$ is the number of energy states below ε calculated in (2.22). In other words, to a first approximation, we assume that the molecules are equally divided among all the levels whose energy is less than ε . By using (2.22) and neglecting numerical factors of order unity, we then have

$$\langle n_r \rangle \sim \frac{N}{\Phi(\varepsilon)} \sim \frac{V}{d^3} \frac{h^3}{V} \frac{1}{(2m\varepsilon)^{3/2}}$$

where $d \sim (V/N)^{1/3} = n^{-1/3}$ is the average distance between molecules. We note that $h/(2m\varepsilon)^{1/2}$ is the de Broglie wavelength of a molecule of energy ε . In anticipation of a result that we shall show very soon, $\varepsilon \sim kT$, we define the *thermal*

³ In fact $\ln Z$ is the generating function of *cumulants* of H (see for example Ma [85] Chapter 12)

$$\langle H^n \rangle_c = \frac{\partial^n \ln Z}{\partial \beta^n}$$

Only for $n = 2$ and $n = 3$ do we have $\langle (\Delta H)^n \rangle = \langle H^n \rangle_c$.

⁴ Mathematically: $\langle n_r \rangle = \text{Tr } D n_r$, where n_r is the particle number operator in state r .

wavelength λ by

$$\lambda = \frac{h}{\sqrt{2\pi mkT}} = \sqrt{\frac{\beta h^2}{2\pi m}} \quad (3.11)$$

where the numerical factor has been chosen in order to simplify subsequent expressions. We can then write

$$\langle n_r \rangle \sim \left(\frac{\lambda}{d} \right)^3 \quad (3.12)$$

When $T \rightarrow \infty$, $\lambda \rightarrow 0$ and $\langle n_r \rangle \rightarrow 0$ at constant density, the probability that a state be doubly occupied is negligible, and so *a priori* are the quantum effects. For example, if the molecules are fermions, the Pauli exclusion principle will have negligible effect under these conditions. The wavelength, λ , is a measure of the ‘quantum size’ of the molecules, and the fact that $\lambda \ll d$ means that the wave packets of different molecules do not overlap. We may thus consider the molecules individually as in classical mechanics. So, the classical approximation will be valid when $\lambda \ll d$. We shall see that in general, in statistical mechanics, a high enough temperature (compared to a given scale) will yield the conditions necessary for this approximation to be valid. In this limit, the partition function for an ideal gas takes the form (2.87)⁵

$$Z_N = \frac{V^N}{N!} \left(\frac{2\pi m}{\beta h^2} \right)^{3N/2} = \frac{1}{N!} \left(\frac{V}{\lambda^3} \right)^N \quad (3.13)$$

⁵ The second expression in (3.13) permits a probabilistic interpretation of the partition function. We can consider V/λ^3 as the number of ‘boxes’, M , in volume V ; M^N is then the number of distinct ways to distribute N distinguishable ‘marbles’ in M labeled boxes with an arbitrary number of marbles per box. We also remark that

$$\langle n_r \rangle \sim \frac{N}{(V/\lambda^3)} = \frac{N}{M}$$

where N/M is indeed the average number of marbles per box. We shall justify in Chapter 5 that classical statistical mechanics is valid in the limit $\langle n_r \rangle \ll 1$. In this limit, we can justify the factor $1/N!$ by making the marbles indistinguishable. By considering the two cases to which Chapter 5 will give physical meaning, the combinatorics is written as

- (i) an arbitrary number of marbles per box ($\equiv$ Bose–Einstein statistics)

$$\frac{(M+N-1)!}{N!(M-1)!} \xrightarrow{N/M \ll 1} \frac{M^N}{N!}$$

- (ii) at most one marble per box ($\equiv$ Fermi–Dirac statistics)

$$\frac{M!}{N!(M-N)!} \xrightarrow{N/M \ll 1} \frac{M^N}{N!}$$

It is straightforward to calculate thermodynamic functions starting with Z_N , or F , if we use the Stirling approximation

$$\ln N! \simeq N \ln N - N = N \ln \frac{N}{e} \quad N \gg 1$$

By introducing the specific volume, $v = V/N$, $\ln Z_N$ becomes

$$\begin{aligned} \ln Z_N &= N \left[\ln \frac{v}{\lambda^3} + 1 \right] \\ &= N \left[\ln \frac{V}{N} + \frac{3}{2} \ln \left(\frac{2\pi m}{\beta h^2} \right) + 1 \right] \end{aligned} \quad (3.14)$$

The average energy is obtained by using (3.2)

$$E = \frac{3N}{2\beta} = \frac{3}{2} NkT \quad (3.15)$$

It is worth noting that the specific heat at constant volume, $C_V = (3/2)Nk$, does not obey the third law. This should not be surprising within the framework of the classical approximation we used here. The free energy $F(T, V, N)$ is given by

$$\begin{aligned} F(T, V, N) &= -NkT \left[\ln \frac{v}{\lambda^3} + 1 \right] \\ &= -NkT \left[\ln \frac{V}{N} + \frac{3}{2} \ln \left(\frac{2\pi mkT}{h^2} \right) + 1 \right] \end{aligned} \quad (3.16)$$

from which we obtain the entropy of the ideal gas

$$S = - \left. \frac{\partial F}{\partial T} \right|_{V,N} = kN \left[\ln \frac{V}{N} + \frac{3}{2} \ln \left(\frac{2\pi mkT}{h^2} \right) + \frac{5}{2} \right] \quad (3.17)$$

This result agrees with that obtained in (2.90b) using the Legendre transform (3.4). The expression for the pressure follows easily

$$P = - \left. \frac{\partial F}{\partial V} \right|_T = \frac{N}{V} kT = nkT$$

where $n = N/V = 1/v$ is the density. This is the 'ideal gas law'. With this expression for the pressure along with Equation (3.15), we verify Equation (2.28) $PV = (2/3)E$. Furthermore, we note that the arbitrary constant $S(V_0, T_0)$ of Problem 1.7.1, which is indeterminate in thermodynamics, is now fixed. We give also the expression for the chemical potential of an ideal gas

$$\mu = \left. \frac{\partial F}{\partial N} \right|_{T,V} = -kT \ln \left(\frac{V}{N} \left(\frac{2\pi mkT}{h^2} \right)^{3/2} \right) = -kT \ln \left(\frac{v}{\lambda^3} \right) = kT \ln(n\lambda^3) \quad (3.18)$$

The chemical potential is negative if $v > \lambda^3$. In fact, this condition is always satisfied because the classical approximation that we have made assumes $v \gg \lambda^3$. We can understand this negative value of μ by starting from $\mu/T = -(\partial S/\partial N)_{E,V}$, which, for large N , is approximately equal to $\mu/T = -(S(N+1) - S(N))$ at constant E and V . If we add a molecule with zero kinetic energy to the system, neither the energy nor the volume of the system will change, but the number of accessible states will increase;⁶ therefore $S(N+1) > S(N)$ and μ is thus negative. When we take into consideration the potential energy due to molecular interactions, the preceding argument is no longer valid because adding a molecule, even if it has zero kinetic energy, will change the energy of the gas. Under these conditions, the chemical potential can become positive, which in fact happens to the ideal Fermi gas but for different reasons (Chapter 5).

Finally, we give a few typical orders of magnitude. The average kinetic energy of a molecule is obtained from the average energy (3.15)

$$\frac{1}{2} m \langle v^2 \rangle = \frac{E}{N} = \frac{3}{2} kT$$

We can then estimate the mean square velocity $\sqrt{\langle v^2 \rangle}$ of a molecule, which we express in units of the velocity of the lightest molecule H_2

$$\sqrt{\langle v^2 \rangle} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{m_{H_2}}{m}} \sqrt{\frac{3kT}{2m_p}} \quad (3.19)$$

where we used $m_{H_2}/m_p \approx 2$ with m_p the proton mass. For a temperature of $T = 300$ K, the numerical value is $\sqrt{\langle v^2 \rangle} \approx 1.93 \times 10^3$ m s⁻¹ for hydrogen, and for other molecules it suffices to divide by $\sqrt{m/m_{H_2}}$. For example, for oxygen, with $m_{O_2} \approx 16m_{H_2}$, we find a speed of 500 m s⁻¹. Under standard temperature and pressure, the molecular volume is 22.4 litres, and the density is $n \approx 2.7 \times 10^{25}$ molecule/m³ which gives the mean inter-molecular distance $d \sim n^{-1/3} \sim 3 \times 10^{-9}$ m. The thermal wavelength of hydrogen at $T = 300$ K is 2×10^{-10} m and it is four times smaller for oxygen: the condition $\lambda \ll d$ is well satisfied.

3.1.3 Paramagnetism

Another very important, and in some cases simple, illustration of the canonical formalism is the study of magnetism. We can relate the macroscopic magnetic properties to the collective behaviour of the microscopic magnetic moments. Some

⁶ An estimate of the number of accessible states is $\Omega_N(E) \approx (\Phi(E/N))^N$ where Φ is given by (2.22). See Exercise 2.7.9 for an exact expression.

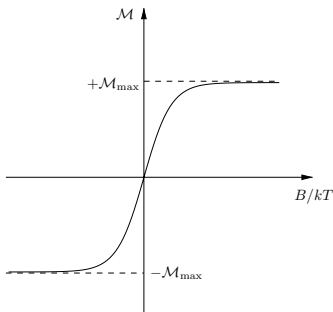


Figure 3.1 Magnetization curve for a paramagnetic material.

materials that do not exhibit spontaneous macroscopic magnetization can be magnetized by subjecting them to an external magnetic field (Figure 3.1). When the macroscopic magnetic moment thus created is aligned with the magnetic field, and when the magnetization does not persist upon removal of the external field, we are dealing with *paramagnetism*. We only describe here the *paramagnetic crystal* model, which is an idealization of a real crystal where we ignore all interactions among the atoms and all degrees of freedom not related to the angular momentum of the atom (or ion). The magnetic field $\vec{B}$ is the external parameter of the problem. Under these conditions, the energy is due entirely to the interaction of the individual moments with the external field. The total Hamiltonian of a system of N magnetic moments is the sum of the individual energies⁷

$$H = \sum_{i=1}^N (-\vec{\mu}_i \cdot \vec{B}) \equiv \sum_{i=1}^N h_i \quad (3.20)$$

Each atom carries a magnetic moment directly proportional to its angular momentum, $\vec{J}$, measured in units of $\hbar$

$$\vec{\mu} = \gamma \vec{J} \quad (3.21)$$

where γ is the *gyromagnetic factor*. The relationship between magnetic moment $\vec{\mu}$ and angular momentum is well known in classical physics. The orbital rotation of

⁷ We use μ for the magnetic moment unless there is confusion with the chemical potential in which case the magnetic moment is denoted by $\bar{\mu}$.

a charge q , subject to a central force, induces a magnetic moment $\vec{\mu} = (q\hbar/2m)\vec{J}$ ($\gamma_{\text{class}} = q\hbar/2m$), where m is the mass of the orbiting particle.⁸ However, classical mechanics does not provide a correct description of magnetism in matter. One must also take into account the magnetism associated with the intrinsic angular momentum of the particles, also called the spin of the particles, the main rôle being played by the electron magnetic moment. The electron has spin-1/2, and the absolute value of its gyromagnetic factor is about twice the *Bohr magneton* $\mu_B = e\hbar/2m_e$, where $-e$ ($e > 0$) and m_e are the charge and mass of the electron.⁹

For illustration, we will take a model system of spin-1/2 where the components of the spin operator $\vec{S}$ (or $\vec{J}$) are proportional to the Pauli matrices σ_i

$$\vec{S} = \frac{1}{2} \vec{\sigma}$$

Even with the simplicity of this model, the calculation of the partition function (3.1b) of the paramagnetic crystal

$$Z = \text{Tr} e^{-\beta H} = \sum_{\{|l\rangle\}} \langle l | e^{-\beta H} | l \rangle \quad (3.22)$$

may appear to be less than obvious due to the very large number (2^N) of possible configurations. But Hamiltonian (3.20) is the sum of independent and identical Hamiltonians, in which case, as we have shown in (2.74) and also used to establish (2.87) for the ideal gas, the partition function factorizes¹⁰

$$Z = \zeta^N \quad \zeta = \text{Tr} e^{-\beta h} \quad (3.23)$$

where ζ is the partition function for a single spin. While the trace of an operator is independent of the basis used, it is simpler to evaluate this trace in the basis of the eigenvectors of the operator, especially if we want the trace of its exponential. It is particularly easy to do this here by choosing the z -axis parallel to the magnetic field, ($\vec{B} = (0, 0, B)$). Then the eigenstates of an individual Hamiltonian are the eigenstates of σ_z , i.e. the states 'spin up', $|+\rangle$, and 'spin down', $|-\rangle$

$$h|\pm\rangle = \mp \mu B |\pm\rangle$$

⁸ Clearly, the appearance of $\hbar$ in γ_{class} is due to our choice of units for $\vec{J}$.

⁹ There is also a magnetic moment due to the nuclear spin whose order of magnitude is given by the nuclear magneton $\mu_N = e\hbar/2m_p$ where m_p is the proton mass: $\mu_B/\mu_N = m_p/m_e \approx 2000$. Its effects are therefore negligible as far as magnetism in matter is concerned. On the other hand, it is the basis of nuclear magnetic resonance (NMR).

¹⁰ There is no factor $1/N!$ as in the case of the ideal gas because the spins are localized to the site of the crystal lattice and can thus be distinguished. If the partition functions are not identical (different B_i or μ_i), $Z = \prod_{i=1}^N \zeta_i$ generalizes (3.23).

In the basis $\{|+\rangle, |-\rangle\}$, the matrix representing the operator $e^{-\beta h}$ is diagonal

$$e^{-\beta h} = \begin{pmatrix} e^{\beta\mu B} & 0 \\ 0 & e^{-\beta\mu B} \end{pmatrix}$$

The partition function for a spin magnetic moment is then

$$\zeta = e^{\beta\mu B} + e^{-\beta\mu B} = 2 \cosh(\beta\mu B) \quad (3.24)$$

Knowing Z , we can use Equation (3.3) to calculate the z -component of the average magnetization in the material, which is, in fact, the conjugate variable of the external field (actually $(-B)$)

$$\mathcal{M} = \frac{1}{\beta} \frac{\partial \ln Z}{\partial B} \Big|_{\beta} = N\mu \tanh(\beta\mu B)$$

The factorization of the partition function (3.23) leads to a simple physical interpretation of the macroscopic magnetization. From Equation (3.24) for the individual partition function, we easily calculate the probability p_+ of finding the spin parallel to the magnetic field (state $|+\rangle$) as well as the probability p_- to find it antiparallel to the field (state $|-\rangle$)

$$p_{\pm} = \frac{1}{\zeta} e^{\pm\beta\mu B} \quad (3.25)$$

The average value of a single magnetic moment along z follows immediately¹¹

$$\langle\mu\rangle = \mu(p_+ - p_-) = \mu \tanh(\beta\mu B) = \mu \tanh\left(\frac{\mu B}{kT}\right) \quad (3.26)$$

The average magnetization of the sample is, then, simply the sum of the average individual moments

$$\mathcal{M} = N\langle\mu\rangle = N\mu \tanh\left(\frac{\mu B}{kT}\right) \quad (3.27)$$

Since the hyperbolic tangent is an increasing function of its argument, we see that the magnetization is a decreasing function of the temperature. When the temperature tends to zero, the spins will increasingly align themselves with the magnetic field, but when the temperature increases, thermal fluctuations tend to destroy this alignment. In fact we are witnessing here an ‘energy–entropy competition’. As we will see in Section 3.5, a system at temperature T will try to minimize its free

¹¹ Clearly, $\langle\mu\rangle$ corresponds to the force conjugate to $(-B)$ for a one-spin system. Equation (3.26) can also be obtained from

$$\langle\mu\rangle = \frac{1}{\beta} \frac{\partial \ln \zeta}{\partial B} \Big|_{\beta}$$

energy $F = E - TS$. At low temperature, it is advantageous to reduce the energy but at high temperature, advantage is gained by increasing the entropy. When the temperature is high enough so that $\mu B \ll kT$, we obtain the Curie law by using the approximation $\tanh x \approx x$, valid for $|x| \rightarrow 0$

$$\mathcal{M} = \frac{N\mu^2 B}{kT} \quad (3.28)$$

Then, at high temperature, the magnetic susceptibility $\chi = \partial \mathcal{M} / \partial B$ takes the value $\mu^2 N / kT$. The susceptibility at zero magnetic field is given, for any temperature, by

$$\chi_0 = \lim_{B \rightarrow 0} \frac{\partial \mathcal{M}}{\partial B} = \frac{N\mu^2}{kT}$$

We have so far only discussed electronic paramagnetism. But, in spite of the small value of the nuclear magneton, nuclear paramagnetism also plays an important rôle. As a widespread application, we mention nuclear magnetic resonance and its offshoots such as medical imaging by magnetic resonance,¹² MRI (Magnetic Resonance Imaging). The central rôle is played by the magnetic moment of the proton which equals 1.41×10^{-26} J/T. In a field of 10 T at normal temperature, $p_+ - p_- \approx \mu B / kT$ is approximately equal to 3×10^{-5} . If we reduce the temperature to that of liquid helium (about 2 K) the orientation effect of the spins becomes much more pronounced: $p_+ - p_- \approx 5 \times 10^{-3}$ (but of course we cannot cool the patients to such convenient temperatures to perform medical imaging!). The interesting effects are obtained by flipping the magnetization, which is done by applying a magnetic field oscillating at the resonance frequency $2\mu B / \hbar$, i.e. 420 MHz for a field of 10 T.

3.1.4 Ferromagnetism and the Ising model

Paramagnetism is a collective exogenous effect: the atomic magnetic moments are totally independent and the magnetization is due solely to the external magnetic field. Certainly, then, a paramagnetic material will not exhibit a magnetization cycle like the one shown in Figure 1.2. That figure shows, on the one hand, that the magnetization is non-zero even when the external magnetic field vanishes and, on the other hand, that it evolves with the magnetic field and traces a hysteresis cycle. This behaviour cannot be explained at all with a Hamiltonian of the form (3.20). Rather, this behaviour is characteristic of a *ferromagnetic* substance. Below a certain critical temperature T_C , called the *Curie temperature* (about 10^3 K for iron),

¹² One should say of course ‘medical imaging by nuclear magnetic resonance’, but the word *nuclear* has been dropped so as not to alarm the public even though this technique has nothing to do with radioactivity. An interesting example of political correctness and marketing savvy.

magnetic substances undergo a second order phase transition from a disordered paramagnetic phase to an ordered ferromagnetic phase characterized notably by the presence of *spontaneous magnetization*¹³ where the magnetization can be non-zero even in the absence of a magnetic field. As for paramagnetism, this magnetization is a result of the alignment of the individual electronic magnetic moments. However, for spontaneous magnetization, this alignment is entirely of an endogenous origin. There is an interaction between the individual electronic magnetic moments which, below the Curie temperature, succeeds in aligning them along the same direction thus forming magnetized domains of a few micrometres. The origin of this interaction is not the direct coupling between the magnetic moments themselves: a simple estimate of the orders of magnitude shows that the Curie temperature would not exceed 1 K in that case. In fact the interaction comes from a combination of the electrostatic repulsion between neighbouring electrons and the antisymmetry of the total wave function. The magnetism of spin-1/2 systems offers a simple example of this 'exchange interaction'.¹⁴ We first consider the spin wave function, which is given by either one of the three states of the spin triplet, which are symmetric under exchange of two spins, or the spin singlet state, which is antisymmetric.¹⁵ The total wave function has to be antisymmetric under the exchange of two electrons, therefore the spatial wave function must be

- (i) antisymmetric when the spins are in a triplet state,
- (ii) symmetric when the spins are in the singlet state.

Ferromagnetism is associated with the triplet, whereas the singlet leads to zero spontaneous magnetization.¹⁶ Nonetheless, the singlet state can lead to an ordered configuration. There exist *antiferromagnetic* materials, which, at a critical temperature called the Néel temperature, undergo a phase transition from a disordered phase to an ordered one corresponding to long range order of alternating spins.

It remains to determine how the selection of one state or another is made. In case (i), the Pauli exclusion principle forbids the electrons to be close to each other, consequently the positive Coulomb potential energy is less important in this

¹³ It can happen that we measure zero spontaneous magnetization in a sample even in its ferromagnetic phase. This is due to cancellation effects between the magnetized domains which form as a result of the phase transition.

¹⁴ This example can be directly applied to magnetism in insulating materials, such as the oxides. Magnetism in conducting materials is more complicated because of the band structure.

¹⁵ The triplet states are given by: $|++\rangle$, $|--\rangle$, $(|+-\rangle + |-+\rangle)/\sqrt{2}$, and the singlet by: $(|+-\rangle - |-+\rangle)/\sqrt{2}$.

¹⁶ A helpful picture that is often used is to consider the magnetic moments in the singlet state as antiparallel vectors whereas they are parallel in the triplet. This is not obvious for the triplet state $(|+-\rangle + |-+\rangle)/\sqrt{2}$, but we can convince ourselves, by writing the rotation matrix of spin 1, that there is a direction $\vec{u}$, such that the triplet can be written as a linear combination of the states $|+\rangle_{\vec{u}}$ and $|-\rangle_{\vec{u}}$ of the operator $\vec{S}_{\text{tot}} \cdot \vec{u}$. Clearly, no rotation permits writing the singlet state in such a form.

case than in case (ii). On the other hand, the kinetic energy here is more important because the momentum is proportional to the gradient of the wave function. A competition between the potential and kinetic energies is thus established. If the former wins, case (i) is chosen, which leads to ferromagnetism, otherwise case (ii) is chosen leading to antiferromagnetism.

To model, in the simplest way possible, the exchange interactions between two electrons on lattice sites i and j of a crystal, we take the following interaction Hamiltonian

$$h_{ij} = -J_{ij} \vec{\sigma}_i \cdot \vec{\sigma}_j$$

where the parameter J_{ij} is taken positive to describe a ferromagnetic interaction and negative for the antiferromagnetic case. By assuming that the interaction strength decreases rapidly enough with distance so that only interactions between nearest neighbours are important, we obtain the Heisenberg Hamiltonian in the presence of an external magnetic field $\vec{B}$

$$H = -\mu \sum_{i=1}^N \vec{\sigma}_i \cdot \vec{B} - J \sum_{\langle i,j \rangle} \vec{\sigma}_i \cdot \vec{\sigma}_j \quad (3.29)$$

where $\langle i, j \rangle$ indicates a sum over only the nearest neighbours. We recognize the first term as the Hamiltonian for paramagnetism. In spite of the apparent simplicity of this Hamiltonian, we do not know how to calculate the partition function exactly except in one dimension.

As a solvable approximation of the Heisenberg Hamiltonian, the *Ising model* has played a big rôle in the study of magnetism.¹⁷ It provoked great advances in the study of the para-ferromagnetic phase transition and contributed to a much deeper understanding of critical phenomena. Our goal here is not an exhaustive presentation of the Ising model, an enormous task due to the huge number of applications it has found even beyond the study of critical phenomena. Our more modest goal is to use an Ising-like model to study the ferromagnetism of spin-1/2, and to give an example of a simple calculation of the partition function when the Hamiltonian is not a simple sum of independent Hamiltonians. Further properties of the Ising model will be examined in Section 4.1.

One of the difficulties of the Heisenberg Hamiltonian is the presence of products of Pauli matrices that do not commute, another is the vector nature of the variables. It is therefore much more convenient to take a model of classical spins, or ‘Ising spins’. An *Ising spin*, S , is a variable that can take the values ± 1 . The classical

¹⁷ The Ising model (actually proposed by Lenz) was solved in one dimension by Ising in 1925 (see Reference [10]). It was necessary then to wait till 1942 for Onsager [97] to give the first exact calculation of the partition function in two dimensions. At the present time, no exact calculation is available in three dimensions.

Ising Hamiltonian corresponding to (3.29) is¹⁸

$$H = -\mu B \sum_{i=1}^N S_i - J \sum_{\langle i,j \rangle} S_i S_j \quad (3.30)$$

The constant, J , is the effective coupling between the spins, the range of this interaction is given by the distance between i and j in $\langle i, j \rangle$. When $J = 0$, Equation (3.30) becomes the Ising version of the Hamiltonian (3.20) which models paramagnetism. The partition function is then

$$Z = \sum_{S_1=\pm 1} \dots \sum_{S_N=\pm 1} \exp \left(\beta J \sum_{\langle i,j \rangle} S_i S_j + \beta \mu B \sum_i S_i \right) \quad (3.31)$$

and contains 2^N terms, one for each possible configuration of the N spins. The trace here becomes a sum over all 2^N possible spin configurations. We can formally express the average total magnetization as

$$\begin{aligned} \mathcal{M} &= \mu \left\langle \sum_i S_i \right\rangle \\ &= \frac{1}{Z} \sum_{S_1=\pm 1} \dots \sum_{S_N=\pm 1} \left(\sum_i S_i \right) \exp \left(\beta J \sum_i S_i S_{i+1} + \beta \mu B \sum_i S_i \right) \quad (3.32) \\ &= \frac{1}{\beta} \frac{\partial \ln Z}{\partial B} \bigg|_{\beta} \end{aligned}$$

Once again, we find the partition function playing the rôle of a generating function.¹⁹

In spite of the great simplification due to the classical approximation, the partition function Z with the Hamiltonian (3.30) can be calculated exactly only in very specific cases: in one dimension, and in two dimensions but with $B = 0$. It is however straightforward to simulate it numerically (see Chapter 7). We shall evaluate the partition function (3.31) in the one-dimensional case with periodic boundary conditions. We thus have a closed geometry shown in Figure 3.2. With these boundary conditions, the partition function can be evaluated exactly using the transfer matrix method which we now outline.

¹⁸ While it is important for the direction of the magnetic field to be uniform in the Ising model, its value can change from site to site. We shall treat this more general situation in Section 4.2.2.

¹⁹ Note that Equation (3.32) has the same form as (2.66). In fact we can treat ferromagnetism by considering the Hamiltonian in zero magnetic field, $H = -J \sum_i S_i S_{i+1}$, and introducing the 'magnetic moment' operator $\mathbf{M}$ ($\mathbf{M} = \mathcal{M}$) as a new relevant operator. The partition function is given by Equation (2.64)

$$Z = \text{Tr} \exp (\lambda_1 H + \lambda_2 \mathbf{M})$$

We can easily show, by considering a quasi-static process, that $\lambda_1 = -\beta$ and $\lambda_2 = \beta \mu B$.

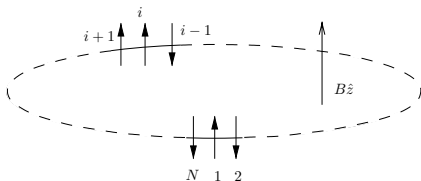


Figure 3.2 One-dimensional system of N Ising spins with periodic boundary conditions.

We first write (3.31) in the form

$$\begin{aligned}
 Z &= \sum_{S_1=\pm 1} \dots \sum_{S_N=\pm 1} \prod_{i=1}^N \exp \left(K S_i S_{i+1} + \frac{K'}{2} (S_i + S_{i+1}) \right) \\
 &= \sum_{S_1=\pm 1} \dots \sum_{S_N=\pm 1} \prod_{i=1}^N T(S_i, S_{i+1})
 \end{aligned} \tag{3.33}$$

with

$$T(S_i, S_j) = \exp \left(K S_i S_j + \frac{K'}{2} (S_i + S_j) \right) \quad \text{and} \quad K = \beta J, \quad K' = \beta \mu B$$

$T(S_i, S_j)$ is a quantity that can take four values determined by the four possible combinations of the two variables S_i and S_j . It can therefore be considered as an element of a 2×2 matrix called the *transfer matrix* that is defined by $\mathcal{T}_{S_i S_j} = T(S_i, S_j)$, i.e.

$$\mathcal{T} = \begin{pmatrix} T(+1, +1) & T(+1, -1) \\ T(-1, +1) & T(-1, -1) \end{pmatrix} = \begin{pmatrix} e^{(K+K')} & e^{-K} \\ e^{-K} & e^{(K-K')} \end{pmatrix}$$

The advantage of this form is in the following equality in which one recognizes a matrix multiplication rule

$$\sum_{S_j=\pm 1} T(S_i, S_j) T(S_j, S_k) = \left(\mathcal{T}^2 \right)_{S_i S_k}$$

and allows us to obtain easily the exact expression of the partition function²⁰

$$\begin{aligned}
 Z &= \sum_{S_1=\pm 1} \left[\sum_{S_2=\pm 1} T(S_1, S_2) \sum_{S_3=\pm 1} \dots \sum_{S_N=\pm 1} T(S_{N-1}, S_N) T(S_N, S_1) \right] \\
 &= \sum_{S_1=\pm 1} \left(T^N \right)_{S_1 S_1} \\
 &= \text{Tr} \left(T^N \right) = \lambda_+^N + \lambda_-^N
 \end{aligned} \tag{3.34}$$

We have used λ_+ and λ_- ($\lambda_+ > \lambda_-$) for the eigenvalues of the transfer matrix

$$\lambda_{\pm} = e^K \left[\cosh K' \pm \sqrt{\sinh^2 K' + e^{-4K}} \right]$$

In the limit $N \gg 1$ we have

$$Z = \lambda_+^N$$

In practice we will use this second expression for Z . Once the partition function is known, we may find expressions for the interesting physical quantities. We can verify that for $J = 0$, Equation (3.32) gives the previous result (3.27) for the magnetization of a paramagnetic substance. We can also evaluate the free energy

$$F = -\frac{N}{\beta} \ln \lambda_+ \tag{3.35}$$

and verify that it is extensive. In the simple case with $B = 0$, the free energy becomes

$$F_0(T, N) = -NkT \ln \left(2 \cosh \frac{J}{kT} \right)$$

We return now to the concept of spontaneous magnetization and its origin. As mentioned earlier, spontaneous magnetization is the result of an energy–entropy competition. The minimization of the free energy $F = E - TS$ requires the best compromise between minimal energy E (all spins aligned) and maximum entropy S (random spins). When the temperature is very low, the entropy plays a smaller rôle, and it is more efficient to minimize E in order to minimize F . Consequently, the spins can become ordered giving a spontaneous magnetization. As the temperature increases, the maximization of the entropy and, therefore, disordered spins are favoured. Thus, thermodynamically, the existence of spontaneous magnetization

²⁰ We propose another method to calculate Z for the simpler case with $B = 0$ in Problem 3.8.1.

is easily understood. However, this reasoning faces a serious difficulty. Examining Equations (3.30) and (3.32) with zero external field, we see clearly that the Hamiltonian is invariant under the symmetry $S_i \rightarrow -S_i$, which leads immediately to the absence of magnetization in zero external field. To apply the above energy–entropy argument, we need to break the symmetry explicitly, for example, by applying an infinitesimal external field.²¹ This forces us to take the thermodynamic limit of the magnetization, more precisely the intensive quantity $\langle m \rangle = \mathcal{M}/N$, which is the magnetization per spin, in the presence of a tiny (positive) external magnetic field, before taking the limit of zero field

$$\langle m \rangle = \lim_{B \rightarrow 0^+} \lim_{N \rightarrow \infty} \frac{1}{N} \mathcal{M} \quad (3.36)$$

The order of the limits is crucial to obtain $\langle m \rangle \neq 0$, i.e. the appearance of spontaneous magnetization.²² If we reverse the order of the limits in (3.36) we obtain $\langle m \rangle = 0$. This situation is typical of *spontaneous symmetry breaking*. In its ferromagnetic phase, the configuration taken by the system has a lower degree of symmetry than the Hamiltonian. The limiting process (3.36) destroys the symmetry $S_i \rightarrow -S_i$. The magnetization per spin, $\langle m \rangle$, which vanishes in the symmetric phase but acquires a non-zero value in the phase with broken symmetry, is called an *order parameter*. Order parameters and symmetry breaking will be discussed further in Section 4.1.

As far as spontaneous magnetization is concerned, the one-dimensional model is special. To see this, it is sufficient to examine the limit (3.36). A lengthy but easy calculation gives

$$\langle m \rangle = \lim_{B \rightarrow 0^+} \lim_{N \rightarrow \infty} kT \left. \frac{\partial \ln \lambda_+}{\partial B} \right|_{\beta} = 0$$

There is no spontaneous magnetization in one-dimension! The absence of symmetry breaking can be understood easily: in one dimension, fluctuations travel the whole system no matter how big it is. There is no sufficiently strong energy constraint (see Problem 3.8.1 for an illustration), and, except for $T = 0$, it is impossible to establish an ordered phase.

²¹ We can also explicitly break the symmetry by aligning all the boundary spins in the same direction and then taking the thermodynamic limit.

²² Obviously

$$\lim_{B \rightarrow 0^-} \lim_{N \rightarrow \infty} \frac{1}{N} \mathcal{M} = -\langle m \rangle$$

3.1.5 Thermodynamic limit

Consider a given extensive quantity²³ $A(T, V, N)$ and assume

$$\lim_{\substack{N, V \rightarrow \infty \\ n \text{ finite}}} \frac{1}{V} A(T, V, N) = a(n, T) \quad (3.37)$$

where $n = N/V$ is the density. We say that the thermodynamic limit of A exists if a has a finite value. In this limit, the physical properties of the system are independent of its size.

In the simple case of an ideal gas, the limit (3.37) of the average energy (3.15) gives a quantity proportional to nkT which is finite for finite n . The energy of an ideal gas is an extensive quantity. Let us now introduce gravity into the problem. The energy acquires a gravitational potential energy term

$$E_G \propto -G \frac{(\rho V)^2}{V^{1/3}} \sim -\rho^2 V^{5/3}$$

where G is the gravitational constant and $\rho = mn$ is the mass per unit volume. When the size of the system increases at constant ρ (or n) we have

$$\frac{E_G}{V} \sim -V^{2/3} \xrightarrow{V \rightarrow \infty} -\infty$$

We see then that the thermodynamic limit no longer exists.²⁴ The dependence on $V^{2/3}$ is a manifestation of the long range nature of the gravitational potential ($\propto 1/r$). The thermodynamic limit is intimately related to the short range nature of interactions. For a system bound by Coulomb interactions, which have infinite range, the existence of the thermodynamic limit is ensured by the presence of charges of opposite signs and screening effects,²⁵ which is not the case for gravitation.

We illustrate the above discussion with the help of the two-dimensional Ising model in zero field.²⁶ Consider a square system of $2N \times 2N = 4N^2$ Ising spins (Figure 3.3(a)). To begin with, we take the unphysical case of identical interactions

²³ The restriction to the variables T, V, N is only for the clarity of presentation. A may depend on any number of variables, extensive or intensive. However, a can be a function only of intensive variables including the ratios of extensive variables.

²⁴ Nonetheless, stars are stable. This is because the gravitational collapse is balanced by pressure if the mass is not too great. However, we do not observe a thermodynamic limit: for example, neutron stars (Exercise 5.6.2) are such that $E \propto M^{-1/3}$. A similar observation applies to white dwarfs (Problem 5.7.3).

²⁵ More precisely, the existence of a thermodynamic limit was proven by Dyson and Lenard [36] under the following three assumptions: (i) the system is globally neutral, (ii) the kinetic energy is given by quantum mechanics, and (iii) at least one of the kinds of charges (positive or negative) is carried by a fermion.

²⁶ Note that for the Ising model the rôle of volume is played by the number of spins.

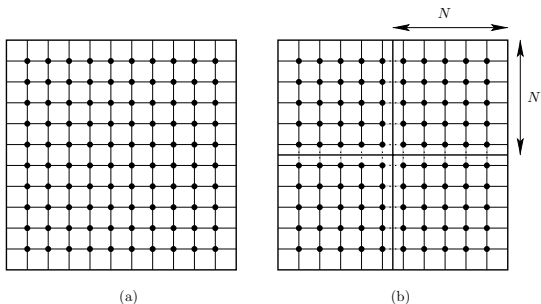


Figure 3.3 (a) Ising system of $4N^2$ spins, and (b) the same system divided into four subsystems of N^2 spins each.

between all the spins, i.e. infinite range interactions. The Hamiltonian becomes

$$H = -J \sum_{i \neq j} S_i S_j \quad (3.38)$$

At $T = 0$, all the spins are aligned and $E \sim N^4$. The limit (3.37) then gives

$$\lim_{N \rightarrow \infty} \frac{E}{N^2} \sim N^2 \rightarrow \infty$$

which shows that the thermodynamic limit does not exist for the Hamiltonian (3.38). Nonetheless, this model is sometimes used as a model for mean field but with an interaction (J/N^2) , which leads to a good thermodynamic limit. We can show that, with this renormalized interaction, the model is exactly solvable and gives the mean field results (Section 4.2).

Now consider the Ising model with interactions only between nearest neighbour spins, and with free boundary conditions

$$H = -J \sum_{\langle i, j \rangle} S_i S_j \quad (3.39)$$

Divide the $4N^2$ spins into four identical subsystems (Figure 3.3(b)), which we assume to be independent. By neglecting the interactions of subsystems at their boundaries, we make an error in the energy ranging between $-4JN$ (boundary spins all parallel) and $4JN$ (boundary spins anti-parallel). On the other hand, the

number of possible spin configurations for the $4N^2$ spins is not modified by the division. These observations can be summarized by

$$(Z_N)^4 e^{-4\beta JN} \leq Z_{2N} \leq (Z_N)^4 e^{4\beta JN} \quad (3.40)$$

where Z_{2N} is the partition function of the complete initial system, Z_N that of one of the $N \times N$ subsystems. We have also used (2.75) to factorize the partition function of the $2N \times 2N$ system composed of four independent $N \times N$ subsystems.

The specific free energies of the problem are therefore

$$\begin{aligned} f_N &= -\frac{1}{\beta} \frac{\ln Z_N}{N^2} && \text{Free energy per spin of an } N \times N \text{ subsystem} \\ f_{2N} &= -\frac{1}{\beta} \frac{\ln Z_{2N}}{4N^2} && \text{Free energy per spin of } 2N \times 2N \text{ system} \end{aligned}$$

By taking the logarithm of (3.40) and dividing by the total 'volume', i.e. $4N^2$, we obtain

$$f_N - \frac{J}{N} \leq f_{2N} \leq f_N + \frac{J}{N}$$

which we write as

$$|f_{2N} - f_N| \leq \text{const} \times \frac{N}{N^2} \sim \frac{L}{\Sigma} \quad (3.41)$$

where $L = N$ is the perimeter of a subsystem and $\Sigma = N^2$ its area. Equation (3.41) shows, on the one hand, that the difference in physical properties between a subsystem and the complete system is due to boundary effects, and, on the other hand, that these boundary effects disappear when $N \rightarrow \infty$. In this limit we can use, with impunity, either the whole system or a part thereof. Equation (3.41) shows that the specific free energies form a Cauchy sequence. They converge to a finite limit, demonstrating again the existence of the thermodynamic limit for the Ising Hamiltonian (3.39). The reader can easily generalize (3.41) to three dimensions by considering a cube of volume $V = 8N^3$ divided into eight identical cubes. The term L/Σ in (3.41) becomes Σ/V .

A similar, but more detailed and complicated argument, allowed Lee and Yang to demonstrate that the free energy of a gas or a fluid is extensive. The crucial point in their derivation is the fact that interactions between molecules are short ranged. In conclusion, the existence of a thermodynamic limit is highly non-trivial and has only relatively recently been demonstrated for a Coulomb gas by Dyson and Lenard [36], see Footnote 25.

3.2 Classical statistical mechanics

3.2.1 Classical limit

It often happens that classical mechanics, which is simpler than quantum mechanics, is sufficient to study a particular physical problem. Such a simplification can also happen in statistical mechanics where classical statistical mechanics may be a good enough approximation. However, one should approach this classical limit with care. We have seen that even in the simple case of an ideal gas, Planck's constant was crucial in fixing the normalization of the entropy and that the identity of the particles (not a classical notion) played a fundamental rôle in ensuring its extensivity. The same is true for the correct expression of the chemical potential. On the other hand, other quantities such as the energy or pressure do not depend on factors of h^{3N} or $N!$ which appear in the partition function (2.87). Except for certain simple cases like Langevin paramagnetism, there is no classical limit, in the strict sense, of statistical mechanics but rather a semi-classical formulation.

Our starting point is the semi-classical equation (2.89) giving the trace for N structureless particles²⁷

$$\text{Tr} \xrightarrow[\text{limit}]{\text{classical}} \frac{1}{N!} \int \frac{\prod_{i=1}^N d^3 p_i d^3 r_i}{h^{3N}} \equiv \frac{1}{N! h^{3N}} \int d\mathbf{p} d\mathbf{q} \quad (3.42)$$

The integration measure, $\prod_i d^3 p_i d^3 r_i / N! h^{3N}$, can be interpreted in the quantum phase space as discussed around Equation (2.88).

Once the trace is defined, all the results we established in the course of the formal construction of the Boltzmann distribution are immediately usable. In particular, $\ln Z$ maintains its rôle as a generating function and, if the classical limit of a quantum dynamical variable $\mathbf{A}$ exists

$$\mathbf{A} \xrightarrow[\text{limit}]{\text{classical}} \mathcal{A}(\mathbf{p}, \mathbf{q})$$

then its average value A can be written as

$$\begin{aligned} A &= \frac{1}{Z} \text{Tr} [\mathbf{A} e^{-\beta H}] \xrightarrow[\text{limit}]{\text{classical}} \frac{1}{Z} \int \frac{d\mathbf{p} d\mathbf{q}}{N! h^{3N}} \mathcal{A}(\mathbf{p}, \mathbf{q}) e^{-\beta H(\mathbf{p}, \mathbf{q})} \\ &= \frac{\int d\mathbf{p} d\mathbf{q} \mathcal{A}(\mathbf{p}, \mathbf{q}) e^{-\beta H(\mathbf{p}, \mathbf{q})}}{\int d\mathbf{p} d\mathbf{q} e^{-\beta H(\mathbf{p}, \mathbf{q})}} \end{aligned} \quad (3.43)$$

The last line of (3.43) shows that in calculating average values, the quantum origin is no longer apparent: neither h nor the combinatoric factor $N!$ appear in the result.

²⁷ We use the following concise notation: $\mathbf{q} = (\vec{r}_1, \dots, \vec{r}_N)$, $\mathbf{p} = (\vec{p}_1, \dots, \vec{p}_N)$, $d\mathbf{q} = d^3\vec{r}_1 \dots d^3\vec{r}_N$ and $d\mathbf{p} = d^3\vec{p}_1 \dots d^3\vec{p}_N$. We consider three-dimensional vectors, but the generalization of the results of Sections 3.2.1 and 3.2.2 to any dimensionality is straightforward.

We have already shown, and we shall verify in more detail in Chapter 5, that the classical limit of statistical thermodynamics corresponds to a situation where the average occupation number of a level is small. For gases, the thermal wavelength gives a measure of the importance of quantum effects with the classical limit given by $\lambda/d \rightarrow 0$.²⁸ Equation (3.12) relates these two views of the classical limit.

3.2.2 Maxwell distribution

Consider the frequently encountered situation where the Hamiltonian separates into kinetic and potential energy terms

$$H(\mathbf{p}, \mathbf{q}) = H_K(\mathbf{p}) + H_U(\mathbf{q}) = \sum_{i=1}^N \frac{\vec{p}_i^2}{2m} + \frac{1}{2} \sum_{i \neq j} U(|\vec{r}_i - \vec{r}_j|) \quad (3.44)$$

In the classical approximation, the term $\exp(-\beta H)$ is a real number called the *Boltzmann weight*. The probability density for a configuration $(\mathbf{p}, \mathbf{q})$ is then given by the normalized Boltzmann weight $Z^{-1} \exp(-\beta H(\mathbf{p}, \mathbf{q}))$. Since the Hamiltonian (3.44) is the sum of two independent terms, the classical partition function factorizes according to (2.74)

$$Z = Z_K Z_U \quad (3.45)$$

It should be emphasized that this factorization is possible only in classical mechanics where the positions and momenta are independent variables. In quantum mechanics, coordinates and momenta are conjugate variables represented by non-commuting operators and consequently it is much more complicated to treat a gas of interacting particles. Z_U is constructed with the potential term in the Hamiltonian

$$Z_U = \frac{1}{V^N} \int d\mathbf{q} e^{-\beta H_U(\mathbf{q})} \quad (3.46)$$

On the other hand, Z_K represents the contribution of the 'kinetic' (translational) degrees of freedom to the partition function. By introducing the thermal wavelength (3.11), Z_K can be written as in (3.13)

$$Z_K = \frac{V^N}{N!} \int \frac{d\mathbf{p}}{h^{3N}} e^{-\beta H_K(\mathbf{p})} = \frac{1}{N!} \left(\frac{V}{\lambda^3} \right)^N \quad (3.47)$$

²⁸ This remark allows us to interpret correctly the entrenched and unfortunate condensed notation for the classical limit: $\hbar \rightarrow 0$. It should be understood to signal the regime where the dimensionless parameter that measures quantum effects becomes negligible.

We recognize in (3.47) the partition function of a classical ideal gas, which is expected since the Hamiltonian of an ideal gas is given by (3.44) but with $U \equiv 0$. For an ideal gas $Z_U = 1$. Once again we note that V/λ^3 is the partition function for a single classical free particle in a box of volume V . Because the momentum and position degrees of freedom are decoupled for a Hamiltonian of the form (3.44), Equation (3.47) shows that to calculate the average value of a function $g(\vec{p}_i)$ that depends only on the momentum of particle i , it is not necessary to know the momenta of the remaining $(N - 1)$ particles.

$$\begin{aligned}\langle g(\vec{p}_i) \rangle &= \frac{\int d\mathbf{p} d\mathbf{q} g(\vec{p}_i) e^{-\beta H(\mathbf{p}, \mathbf{q})}}{\int d\mathbf{p} d\mathbf{q} e^{-\beta H(\mathbf{p}, \mathbf{q})}} = \frac{\int d\mathbf{q} e^{-\beta H_U(\mathbf{q})} \int d\mathbf{p} g(\vec{p}_i) e^{-\beta H_K(\mathbf{p})}}{\int d\mathbf{q} e^{-\beta H_U(\mathbf{q})} \int d\mathbf{p} e^{-\beta H_K(\mathbf{p})}} \\ &= \frac{\int d^3 p_i g(\vec{p}_i) \exp\left[-\frac{\beta}{2m} \vec{p}_i^2\right]}{\int d^3 p_i \exp\left[-\frac{\beta}{2m} \vec{p}_i^2\right]}\end{aligned}$$

Since the particles are identical, the final result does not depend on the particle considered

$$\langle g(\vec{p}_i) \rangle \equiv \langle g(\vec{p}) \rangle = \frac{1}{(2\pi mkT)^{3/2}} \int d^3 p g(\vec{p}) \exp\left[-\frac{\vec{p}^2}{2mkT}\right] = \int d^3 p g(\vec{p}) \bar{\varphi}(\vec{p}) \quad (3.48)$$

The last equation in (3.48) defines the *Maxwell distribution*²⁹ for the momenta, $\bar{\varphi}(\vec{p})$, or the velocities, $\varphi(\vec{v})$

$$\bar{\varphi}(\vec{p}) \hat{=} \frac{1}{(2\pi mkT)^{3/2}} \exp\left[-\frac{\vec{p}^2}{2mkT}\right] = \frac{1}{m^3} \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left[-\frac{m\vec{v}^2}{2kT}\right] \hat{=} \frac{1}{m^3} \varphi(\vec{v})$$

(3.49)

$\bar{\varphi}(\vec{p}) d^3 p = \varphi(\vec{v}) d^3 v$ is the probability of finding a particle in the volume $d^3 p$ around $\vec{p}$ (or $d^3 v$ around $\vec{v}$). We emphasize that the Maxwell distribution is general and always valid, as long as quantum effects are negligible, no matter what correlations are introduced by H_U .

The distributions (3.49) factorize into the products of three Gaussian distributions for the three independent spatial directions. They are centred at zero with variance mkT for $\bar{\varphi}(\vec{p})$ or kT/m for $\varphi(\vec{v})$

$$\langle p_\alpha^2 \rangle = mkT \qquad \langle v_\alpha^2 \rangle = \frac{kT}{m} \quad (3.50)$$

²⁹ It is clear, for obvious chronological reasons, that the 'post-Boltzmannian' reasoning we follow here, and which represents the Maxwell distribution as a simple consequence of the canonical equilibrium distribution, is not the reasoning followed by Maxwell who developed his theory in 1860.

We immediately conclude that the average kinetic energy per particle is

$$\left\langle \frac{1}{2} m \vec{v}^2 \right\rangle = 3 \left\langle \frac{1}{2} m v_\alpha^2 \right\rangle = \frac{3}{2} kT \quad (3.51)$$

The average kinetic energy of the system, i.e. the average total energy for an ideal gas, is obtained by a sum of the N identical contributions (3.51), in agreement with (3.15)

$$E = \frac{3}{2} N kT \quad (3.52)$$

We shall see in the following section that these last two results are special cases of the equipartition theorem.

From the average of the kinetic energy (3.51), we can define a typical thermal speed $\sqrt{\langle v^2 \rangle}$. We say ‘a typical’ and not ‘the typical’ speed because there are other ‘typical’ speeds. To understand the origin of this, consider the distribution of the absolute values of the velocities: it is not a Gaussian!

$$\begin{aligned} \phi(v) dv &= \int d\Omega v^2 dv \varphi(\vec{v}) \\ &= 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} v^2 \exp \left[-\frac{m \vec{v}^2}{2kT} \right] dv \end{aligned} \quad (3.53)$$

This distribution law is called a *Maxwellian* and is plotted in Figure 3.4. In addition to the typical speed obtained from the equipartition theorem, we also show in this

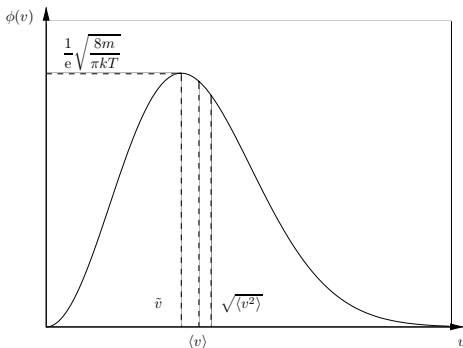


Figure 3.4 Distribution of the velocity modulus.

figure the most probable speed $\tilde{v}$ which corresponds to the maximum of the probability density, and the average speed $\langle v \rangle$. These three speeds are given by

$$\tilde{v} = \sqrt{\frac{2kT}{m}} \quad (3.54a)$$

$$\langle v \rangle = \sqrt{\frac{8kT}{\pi m}} = 1.13 \times \tilde{v} \quad (3.54b)$$

$$\sqrt{\langle v^2 \rangle} = \sqrt{\frac{3kT}{m}} = 1.22 \times \tilde{v} \quad (3.54c)$$

The small differences in the numerical values show that for an estimate of orders of magnitude, we can equally well use one or another.

3.2.3 Equipartition theorem

The result in Equation (3.51), and the large number of its experimental verifications, have greatly contributed to the emergence of the statistical picture of thermodynamics. It is an illustration of a major theorem of classical statistical mechanics called the *equipartition theorem*. The validity and application of this theorem extend far beyond the simple framework of physical situations described by a Hamiltonian of the form (3.44). We shall establish it for a Hamiltonian of the following general form

$$H = \sum_{i,j=1}^{\nu} x_i a_{ij} x_j + \tilde{H} \quad (3.55)$$

where x_i represents a component of a coordinate or momentum vector. The first term on the right hand side is the ‘quadratic part’ of the Hamiltonian where a_{ij} is a constant element of the $\nu \times \nu$ symmetric and strictly positive matrix a .³⁰ The integer parameter ν , $\nu \leq 2M$, where M is the number of degrees of freedom, counts the number of ‘quadratic coordinates’ of the system. The other part of the Hamiltonian, $\tilde{H}$, is the non-quadratic part, and regroups all the contributions that cannot be written under that form. It depends *only* on the $(2M - \nu)$ coordinates other than the x_i .

Let us calculate the average value of the quadratic part of the Hamiltonian

$$\left\langle \sum_{i,j=1}^{\nu} x_i a_{ij} x_j \right\rangle = \sum_{i,j=1}^{\nu} a_{ij} \langle x_i x_j \rangle$$

³⁰ The a_{ij} can be functions of x_k for $k > \nu$ (see (3.60)). We leave this generalization of the theorem as an exercise for the reader.

All average values can be calculated with the total Hamiltonian by using (3.43), but here the contribution of $\tilde{H}$ drops out. This is precisely the simplification that has been used to establish (3.48). The evaluation of the average value $\langle x_i x_j \rangle$ involves only Gaussian integrals, a result which we give in (A.45), with $A_{ij} = 2\beta a_{ij}$

$$\langle x_i x_j \rangle = \frac{kT}{2} (a^{-1})_{ij}$$

We thus establish the equipartition theorem which stipulates that each quadratic degree of freedom contribute $kT/2$ to the average value of the Hamiltonian

$$\langle H \rangle = \nu \frac{kT}{2} + \langle \tilde{H} \rangle \quad (3.56)$$

To illustrate this theorem we take the simple but physically important case of a particle in a one-dimensional harmonic potential

$$H(p, x) = \frac{p^2}{2m} + \frac{1}{2} \kappa x^2$$

The theorem (3.55) gives

$$\left\langle \frac{p^2}{2m} \right\rangle = \left\langle \frac{1}{2} \kappa x^2 \right\rangle = \frac{1}{2} kT \quad (3.57)$$

We remark that the first equality in (3.57) is the virial theorem of classical mechanics: the equality, on average, of the potential and kinetic energies. For our Hamiltonian that has two quadratic coordinates we have

$$E = kT \quad C_V = k \quad (3.58)$$

We see with this example one of the important consequences of the equipartition theorem: knowing the number ν of quadratic coordinates immediately allows us to calculate their contribution to the specific heat, $C_V = \nu k$. We shall see in the next section how including the quantum effects at the microscopic scale imposes limits on the equipartition theorem. In particular, we shall be led to define the notion of ‘frozen degrees of freedom’.

To conclude, note that in the classical limit, fluctuations of the energy, just like its average value, are determined by kT . In fact, for a purely quadratic Hamiltonian, the fluctuation-response theorem (3.10) becomes, with the help of (A.46),

$$\langle (\Delta H)^2 \rangle = \frac{\nu}{2} k^2 T^2 \quad (3.59)$$

which, according to (3.10), is in agreement with $C_V = \nu k/2$.

3.2.4 Specific heat of a diatomic ideal gas

We now apply the equipartition theorem to an ideal gas of N diatomic molecules. Since the assumptions behind the ideal gas model ensure that the partition function factorizes, it only remains to identify correctly the various degrees of freedom of a molecule. In addition to translational motion, which brings in three degrees of freedom, a diatomic molecule can have rotational and vibrational motion which we take to be independent in first approximation. To model the rotation we assume that the molecule is made of two atoms, masses m_1 and m_2 , forming a dumb-bell that can turn around an axis of rotation perpendicular to the line joining the two nuclei, and whose moment of inertia is I (Figure 3.5). Using the angles defined in Figure 3.5, the rotation Lagrangian for a single molecule can be written as

$$\mathcal{L}_{\text{rot}} = \frac{1}{2}I(\dot{\theta}^2 + \sin^2 \theta \dot{\varphi}^2)$$

which leads to the Hamiltonian

$$h_{\text{rot}} = \frac{1}{2I} \left(p_{\theta}^2 + \frac{1}{\sin^2 \theta} p_{\varphi}^2 \right) \quad (3.60)$$

We have introduced the generalized momenta p_{θ} and p_{φ} which are defined by

$$p_{\theta} = \frac{\partial \mathcal{L}_{\text{rot}}}{\partial \dot{\theta}} = I \dot{\theta} \quad p_{\varphi} = \frac{\partial \mathcal{L}_{\text{rot}}}{\partial \dot{\varphi}} = I \sin^2 \theta \dot{\varphi}$$

According to (3.60), we must associate two degrees of freedom and two quadratic coordinates ($\nu = 2$ in (3.56)) with rotational motion.

The (small) vibrational motion of a molecule can be interpreted as the harmonic oscillations of an effective particle whose mass is the reduced mass,

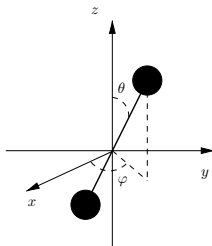


Figure 3.5 A diatomic molecule.

$\mu = m_1 m_2 / (m_1 + m_2)$, and whose normal mode frequency, ω_{vib} , depends on the molecule. The corresponding Hamiltonian can be written as

$$h_{\text{vib}} = \frac{p^2}{2\mu} + \frac{1}{2}\mu \omega_{\text{vib}}^2 q^2 \quad (3.61)$$

The system has one degree of freedom and two quadratic coordinates.

We can now count the number of quadratic coordinates for an ideal diatomic gas and apply the equipartition theorem to calculate its specific heat

$$\left. \begin{array}{ll} E_{\text{tr}} = \frac{3}{2}NkT & C_V^{\text{tr}} = \frac{3}{2}Nk \\ E_{\text{rot}} = NkT & C_V^{\text{rot}} = Nk \\ E_{\text{vib}} = NkT & C_V^{\text{vib}} = Nk \end{array} \right\} \Rightarrow C_V = \frac{7}{2}Nk \quad (3.62)$$

Figure 3.6 shows the generic behaviour of the specific heat of an ideal diatomic gas. The equipartition theorem appears to be satisfied only at ‘high temperature’. In this problem with no reference scales, the ambiguity of the notion of high temperature reflects limitations of the classical description. As the temperature decreases, we observe big deviations from the classical predictions. Whereas the onset of the deviations is determined by the temperature, their values are not. Furthermore, these deviations are well beyond what might be explained by anharmonic effects. However, the plateaux shown in the figure, at $5/2Nk$ and $3/2Nk$, correspond to values that can be explained by the classical approach on condition that we ‘forget’ certain types of motion and their degrees of freedom. The next section will justify this piece-wise validity of the equipartition theorem by taking into account quantum effects and in particular the discrete nature of the energy levels associated with different kinds of motion. It is this notion of degrees of freedom that justifies taking only rotations perpendicular to the symmetry axis of the molecule and not rotations around that axis. In fact, the moment of inertia in the second case is very small and the corresponding rotational energies extremely high.

3.3 Quantum oscillators and rotators

3.3.1 Qualitative discussion

To understand Figure 3.6 we re-examine the structure of a diatomic molecule. We consider a molecule made of two atoms, with masses m_1 and m_2 (atomic masses A_1 and A_2). The Born–Oppenheimer approximation exploits the big difference in mass between the nuclei and the electrons to consider the former immobile and take into account only the motion of the electrons in the reference frame where the nuclei are at rest. To determine the potential that binds the atoms into a molecule, one follows the evolution of the electronic wave functions, as a function of nuclear

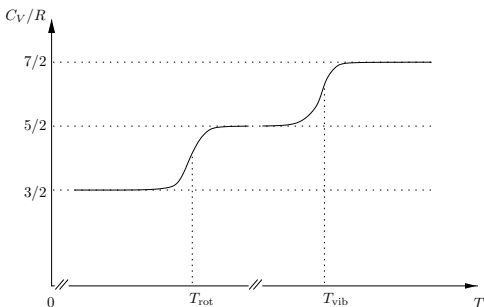


Figure 3.6 The specific heat of a mole of ideal gas as a function of the temperature.

separation. For a large majority of molecules in their ground state, the projection of the total orbital angular momentum along the symmetry axis is zero. The only control parameter is the distance R between the two nuclei. For each value of R , we determine the electronic wave function, which in turn determines the intensity of the interaction between the two nuclei. In this way we determine, step by step, the potential energy $U(R)$ of the molecule in its ground state. The typical qualitative form of $U(R)$ is shown in Figure 3.7.

The potential energy $U(R)$ has the form of a Lennard-Jones potential similar to what is used to describe interaction between molecules. However, the characteristic scales, R_0 and E_0 , are very different in the two systems, so this should not lead to any confusion. The exact form of the potential depends a great deal on the molecule under consideration. For order-of-magnitude estimates, we can take as the position of the minimum the radius of the Bohr atom ($R_0 \simeq a_0 \simeq 0.053 \text{ nm}$) and for the depth of the well the Rydberg ($|E_0| \simeq \text{Ry} \simeq 13.6 \text{ eV}$). With R_0 and the reduced mass we can calculate the moment of inertia

$$I \simeq \mu a_0^2$$

To estimate the order of magnitude of a typical vibrational energy, we first state the obvious: a diatomic molecule can only exist if the energies associated with its internal degrees of freedom are much smaller than its dissociation energy. The vibrational levels are therefore at the bottom of the potential well, and, as such, are

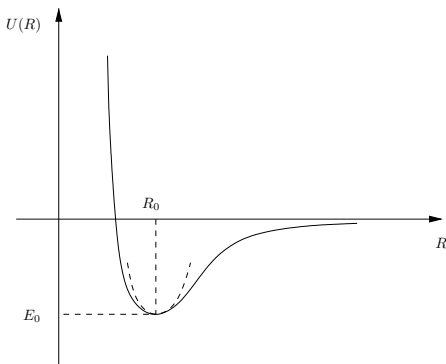


Figure 3.7 The general form of the molecular potential.

very well described by its parabolic approximation (see Figure 3.7)

$$U(R) \simeq U(R_0) + \frac{1}{2}\kappa(R - R_0)^2$$

with

$$\kappa = \left. \frac{d^2U}{dR^2} \right|_{R=R_0} \simeq \frac{\text{Ry}}{a_0^2}$$

We are now ready to calculate the characteristic rotational and vibrational energies, and the relevant temperature scales

$$\bar{\varepsilon}_{\text{vib}} = \hbar\omega_{\text{vib}} = \hbar\sqrt{\frac{\kappa}{\mu}} \simeq \frac{1}{\sqrt{\alpha}}\sqrt{\frac{m_e}{m_p}}\text{Ry} \quad T_{\text{vib}} = \bar{\varepsilon}_{\text{vib}}/k \simeq \frac{1}{\sqrt{\alpha}} \times 3700 \text{ K} \quad (3.63a)$$

$$\bar{\varepsilon}_{\text{rot}} = \frac{\hbar^2}{2I} \simeq \frac{1}{\alpha} \frac{m_e}{m_p} \text{Ry} \quad T_{\text{rot}} = \bar{\varepsilon}_{\text{rot}}/k \simeq \frac{1}{\alpha} \times 85 \text{ K} \quad (3.63b)$$

where α represents the reduced mass in units of the proton mass

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \simeq \frac{A_1 A_2}{A_1 + A_2} m_p = \alpha m_p$$

Let us examine again Figure 3.6. At high temperature, i.e. $T_{\text{vib,rot}} \ll T$, all degrees of freedom are excited, and since the number of accessible states is huge, the classical approximation is valid. Recall that for a classical particle the average energy and the amplitude of fluctuations are determined by kT . When the temperature, which is the only source of energy for the molecule, goes below the temperature associated with a particular motion, that motion can no longer be excited. We thus speak of ‘frozen’ degrees of freedom where the only occupied level is the ground state, a situation very far from the regime of validity of the classical approximation. In Figure 3.6 we first observe the freezing of the vibrational and then the rotational degrees of freedom. We note that for many gases it is impossible to observe the freezing of the rotational degrees of freedom because the required temperature is below the liquefaction temperature of the gas itself, a situation where the ideal gas model is obviously no longer valid for a liquid. It might appear surprising that the specific heat does not vanish at zero temperature as required by the third law. However, the model presented here is not complete since it does not take into account the effects of quantum statistics.

3.3.2 Partition function of a diatomic molecule

To make precise the notion of frozen degrees of freedom, we calculate explicitly the vibrational and rotational partition functions for a diatomic molecule.³¹ It is sufficient to calculate the individual partition functions because the different kinds of degrees of freedom are independent and, consequently, the partition function for an ideal gas of diatomic molecules can be written as

$$Z_N = \frac{1}{N!} (\zeta(\beta) \zeta_{\text{vib}} \zeta_{\text{rot}})^N \quad (3.64)$$

where $\zeta(\beta) = V/\lambda^3$ is the translational partition function (see (3.47)) for one molecule. The evaluation of ζ_{vib} and ζ_{rot} is the object of this section.

With the harmonic approximation of the potential well, the vibrational energy levels are those of the quantum simple harmonic oscillator

$$\varepsilon_n^{\text{vib}} = -u_0 + \left(n + \frac{1}{2}\right) \hbar \omega_{\text{vib}} \quad (3.65)$$

³¹ We will limit ourselves to molecules composed of two different molecules (heteronuclear). For homonuclear molecules we have the additional requirement of making the total wave function antisymmetric under the exchange of the two nuclei. Problem 3.8.3 treats this situation with the example of the hydrogen molecule.

with $u_0 = -E_0 \simeq \text{Ry}$. The evaluation of the partition function is therefore reduced to summing a geometric series

$$\zeta_{\text{vib}} = \sum_{n=0}^{\infty} e^{-\beta \varepsilon_n^{\text{vib}}} = \exp[\beta u_0] \frac{\exp[-\beta(\hbar\omega_{\text{vib}}/2)]}{1 - \exp[-\beta\hbar\omega_{\text{vib}}]} \quad (3.66)$$

The average vibrational energy is given by

$$\varepsilon_{\text{vib}} = -\frac{\partial \ln \zeta_{\text{vib}}}{\partial \beta} = -u_0 + \left(\frac{1}{\exp[\beta\hbar\omega_{\text{vib}}] - 1} + \frac{1}{2} \right) \hbar\omega_{\text{vib}} \quad (3.67)$$

Comparing (3.65) and (3.67) we find the average thermally excited energy level of the harmonic oscillator $\langle n \rangle = (\exp[\beta\hbar\omega_{\text{vib}}] - 1)^{-1}$. We will return to the physical interpretation of this quantity in Section 5.4.

The specific heat³² is then given by

$$C_V = k \left(\frac{T_{\text{vib}}}{T} \right)^2 \frac{e^{T_{\text{vib}}/T}}{(e^{T_{\text{vib}}/T} - 1)^2} \simeq \begin{cases} k (T_{\text{vib}}/T)^2 e^{-T_{\text{vib}}/T} \rightarrow 0 & \text{for } T \ll T_{\text{vib}} \\ k & \text{for } T \gg T_{\text{vib}} \end{cases} \quad (3.68)$$

When $T \gg T_{\text{vib}}$, i.e. when the available energy, kT , is large compared to the level spacing in the harmonic oscillator, $\hbar\omega_{\text{vib}}$, the classical approximation is valid and we regain the prediction of the equipartition theorem. At the other end, $T \ll T_{\text{vib}}$, we verify that this equation is in agreement with the picture of frozen degrees of freedom, and the specific heat vanishes.³³

The rotational states are described by spherical harmonics, $Y_{jm}(\theta, \varphi)$, which are simultaneous eigenstates of J^2 (eigenvalues $j(j+1)$) and J_z (eigenvalues m). In the Born–Oppenheimer model, the rotational levels are independent of m , and are consequently $(2j+1)$ -fold degenerate,

$$\varepsilon_{j,m}^{\text{rot}} = j(j+1) \frac{\hbar^2}{2I} = j(j+1) kT_{\text{rot}} \quad (3.69)$$

³² The specific heat of a quantum oscillator is of great importance in the history of science because it corresponds to that of a solid (up to an overall numerical factor) in the model proposed by Einstein (Exercise 3.7.2). By successfully describing the vanishing of the specific heat of solids at very low temperatures, observed experimentally but not explained by classical physics (the Dulong–Petit law, based on the classical equipartition theorem, predicts a constant specific heat), it reinforced the power of the newly discovered quantum theory. However, Einstein's model does not describe vibrations in solids satisfactorily: the reader is referred to Section 5.4 for a more satisfactory model due to Debye. Einstein's model proposes $\omega_k = \text{const}$ in (5.47), which is equivalent to assuming that all atoms vibrate at the same frequency.

³³ The exponential decrease of the specific heat in Einstein's model is to be compared with the power law T^3 found in (5.64) and which is observed experimentally.

The rotational partition function, then, takes the following form

$$\zeta_{\text{rot}} = \sum_{j=0}^{+\infty} \sum_{m=-j}^{+j} e^{-\beta \varepsilon_{j,m}^{\text{rot}}} = \sum_{j=0}^{+\infty} (2j+1) \exp \left[-j(j+1) \frac{T_{\text{rot}}}{T} \right] \quad (3.70)$$

For most molecules (see (3.63b)) $T_{\text{rot}} \lesssim 10$ K, there is therefore a large range of temperatures such that $T_{\text{rot}}/T \ll 1$. In this regime, the discrete structure of the rotational energy levels is totally washed out, $kT \gg \bar{\varepsilon}_{\text{rot}}$ and we can adopt the classical description. This allows us to replace the discrete sum in (3.70) with an integral

$$\zeta_{\text{rot}} \simeq \int_0^{+\infty} dx (2x+1) \exp \left[-x(x+1) \frac{T_{\text{rot}}}{T} \right] \simeq \frac{T}{T_{\text{rot}}} \quad (3.71)$$

This expression for the partition function was obtained in the classical approximation. It is easy to verify that the average energy and the specific heat obtained from it are in agreement with the equipartition theorem.

$$\varepsilon_{\text{rot}} = -\frac{\partial}{\partial \beta} \ln \zeta_{\text{rot}} = kT \quad \text{and} \quad C_V = \frac{\partial \varepsilon_{\text{rot}}}{\partial T} = k$$

For temperatures below a few tens of kelvins, or even for higher temperature but for light molecules such as H_2 or HD, the classical approximation is no longer valid. We therefore enter the non-classical regime where $T_{\text{rot}}/T \gg 1$ and for which the rotation of molecules is not favoured energetically. Under these conditions, only the first few terms of the sum (3.70) will contribute significantly to the partition function

$$\zeta_{\text{rot}} = 1 + 3e^{-2T_{\text{rot}}/T} + 5e^{-6T_{\text{rot}}/T} + \dots \simeq 1$$

The exponential behaviour of the specific heat, for $T \rightarrow 0$, observed in (3.68) for the harmonic oscillator and which also holds for rotational motion, is *characteristic of an energy gap, ΔE , between the ground state and the first excited state*

$$C \sim \exp \left(-\frac{\Delta E}{kT} \right) \quad (3.72)$$

3.4 From ideal gases to liquids

The thermodynamics of a gas at low density and high temperature, where interactions between atoms are negligible, can be accurately calculated using the ideal gas model whose partition function is easy to evaluate. We shall also see, in Chapter 5,

that at lower temperatures where quantum effects can no longer be ignored, we can construct a quantum ideal gas model. Somewhat surprisingly, these two models can often be generalized to density and temperature regimes characteristic of solids where correlations among constituent particles are very strong. In fact, in favourable cases, the statistical thermodynamics of solids reduces to ideal gas models of collective excitations that give rise in the quantum case to 'quasi-particles'. These quasi-particles, which are often independent to first approximation, are the quanta of collective propagation modes. For example, the interactions between the electrons and the lattice potential in a metal lead to the introduction of the concept of conduction electrons. These are effective, approximately independent quasi-particles whose properties, such as mass, are different from those of free electrons. Their statistical properties, obtained from a quantum ideal gas model, explain the low temperature specific heat in metals (see Section 5.2). In the same way, the specific heat of insulating solids is explained by considering a quantum ideal gas of 'phonons' which are quasi-particles corresponding to the vibration modes of the crystal lattice (see Section 5.4).

What is the situation for liquids? To begin with, it is interesting and instructive to ask what is the basic property of the liquid state. This is an important question because we quickly realize that it is impossible to bring out a basic difference in the behaviour of liquids and gases. Only the presence of an interface between the two states allows us to distinguish immediately between them. This ambiguity is clear when we examine a typical phase diagram such as shown in Figure 3.8. We see from these diagrams that it is possible to go around the critical point, thus passing continuously from the liquid to the gaseous phase, which is not true for the liquid–solid and gas–solid transitions. We also encounter great difficulty in trying to establish a difference between a liquid and an amorphous substance. Briefly, we can say that any substance that is neither a gas nor a crystalline solid is a

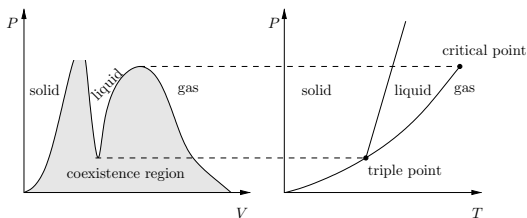


Figure 3.8 The $P-V$ and $P-T$ phase diagrams of a representative material.

liquid. In an ideal gas the molecules are totally independent while in a solid there is *long range order*. In a liquid, no such long range order is present which provides a qualitative difference with solids. However, there are *short range correlations*, which are much stronger in a liquid than in a gas. In the following paragraphs, we will demonstrate how these correlations determine the thermodynamics of liquids and dense gases.

3.4.1 Pair correlation function

We work in the framework of the classical approximation and take a Hamiltonian of the form (3.44). The partition function factorizes according to (3.45) into an 'ideal gas' part

$$Z_K = \frac{1}{N!} \left(\frac{V}{\lambda^3} \right)^N$$

and a potential energy term which controls the dependence on spatial coordinates

$$Z_U = \frac{1}{V^N} \int \prod_{i=1}^N d^3 r_i e^{-\beta H_U} \quad H_U = \frac{1}{2} \sum_{i \neq j} U(\vec{r}_i, \vec{r}_j)$$

The integration measure $\prod d^3 r_i$ represents a volume element in the $3N$ dimensional space formed by the N position vectors. The statistical weight of this volume element depends on its spatial position. The joint probability to find particle 1 in element $d^3 r_1$ at $\vec{r}_1$, particle 2 in $d^3 r_2$ at $\vec{r}_2$... particle N in $d^3 r_N$ at $\vec{r}_N$ is given by the normalized Boltzmann weight

$$\mathcal{P}_N(\vec{r}_1, \dots, \vec{r}_N) \frac{d^3 r_1 \dots d^3 r_N}{V^N} = \frac{e^{-\beta H_U(\vec{r}_1, \dots, \vec{r}_N)}}{Z_U} \frac{d^3 r_1 \dots d^3 r_N}{V^N} \quad (3.73)$$

The probability, $\mathcal{P}_1(\vec{r}_i) d^3 r_i$, of finding particle i in the neighbourhood of $\vec{r}_i$ for any position of the other $N - 1$ particles is obtained by integrating over all coordinates $j \neq i$

$$\mathcal{P}_1(\vec{r}_i) d^3 r_i = \frac{d^3 r_i}{V} \int_V \dots \int_V \frac{\prod_{j \neq i} d^3 r_j}{V^{N-1}} \mathcal{P}_N(\vec{r}_1, \dots, \vec{r}_N) \quad (3.74)$$

We see that probability distributions are not very convenient since, for example, $\mathcal{P}_1(\vec{r}_i) \sim 1/V \ll 1$. A more direct, but totally equivalent approach, is to choose a region of space and count the average number of molecules it contains.

In this approach we deal with local densities. The simplest among them is the number density, i.e. the number of particles per unit volume

$$n(\vec{r}) = \frac{1}{Z_U} \int_V \dots \int_V \frac{d^3 r_1 \dots d^3 r_N}{V^N} \sum_{i=1}^N \delta(\vec{r} - \vec{r}_i) e^{-\beta H_U} = \left\langle \sum_{i=1}^N \delta(\vec{r} - \vec{r}_i) \right\rangle \quad (3.75)$$

The quantity

$$n(\vec{r}) d^3 r = \left\langle \sum_{i=1}^N \delta(\vec{r} - \vec{r}_i) \right\rangle d^3 r$$

is the average number of particles in a volume $d^3 r$ at $\vec{r}$. Clearly, n is directly related to $\mathcal{P}_1$:³⁴ $n(\vec{r}) = N \mathcal{P}_1(\vec{r})$.

Similarly we define the *pair density* as follows

$$n_2(\vec{r}, \vec{r}') \equiv \left\langle \sum_{i=1}^N \sum_{j \neq i} \delta(\vec{r} - \vec{r}_i) \delta(\vec{r}' - \vec{r}_j) \right\rangle \quad (3.76)$$

where $n_2(\vec{r}, \vec{r}') d^3 r d^3 r'$ is the joint average number of pairs of particles at $\vec{r}$ and $\vec{r}'$ and the average value $\langle \bullet \rangle$ is defined in (3.75). We can easily show that this definition of the pair density satisfies the following expected result

$$\int_V d^3 r' n_2(\vec{r}, \vec{r}') = (N - 1) n(\vec{r})$$

Indeed a given particle has $(N - 1)$ chances of pairing up. When the density is uniform, which is the case in the absence of an external potential, we have $n(\vec{r}) \equiv n = N/V$. Then, for $N \gg 1$,

$$\frac{1}{V} \int_V d^3 r' n_2(\vec{r}, \vec{r}') \simeq n^2$$

We introduce in this case the *pair correlation function*, $g(\vec{r}, \vec{r}')$, which measures the degree of correlation among the particles³⁵

$$n_2(\vec{r}, \vec{r}') = n^2 g(\vec{r}, \vec{r}') \quad (3.77)$$

³⁴ $n(\vec{r}) = \sum_{i=1}^N \int_V d^3 r_i \delta(\vec{r} - \vec{r}_i) \mathcal{P}_1(\vec{r}_i)$, and for one type of particle all the probabilities are identical.

³⁵ If the density is not uniform, we define $g(\vec{r}, \vec{r}')$ by

$$g(\vec{r}, \vec{r}') = \frac{V}{(N - 1)n(\vec{r})} n_2(\vec{r}, \vec{r}')$$

Note that $g(\vec{r}, \vec{r}') d^3r' / V$ can be interpreted as the conditional probability of finding a particle in volume d^3r' at $\vec{r}'$ knowing there is a particle at $\vec{r}$. By definition, $g(\vec{r}, \vec{r}') = 1$ for an ideal gas where gas particles are totally independent (no correlations). In the general case, g takes the asymptotic value $g = 1$, even in a dense gas, because as the separation between particles increases, their correlation disappears

$$\lim_{|\vec{r}-\vec{r}'| \rightarrow \infty} g(\vec{r}, \vec{r}') = 1 \quad (3.78)$$

Very often, the fluid properties are uniform and isotropic. This is the case when the potential and the correlation function depend neither on position nor on the relative orientation of the two particles: $U(\vec{r}_i, \vec{r}_j) = U(|\vec{r}_i - \vec{r}_j|)$ and $g(\vec{r}, \vec{r}') = g(|\vec{r} - \vec{r}'|) \equiv g(r)$.

The pair correlation function is a very important quantity because, on the one hand, it can be measured experimentally (Section 3.4.2) and, on the other hand, it is related to important thermodynamic quantities like energy and pressure (Section 3.4.3). We can easily get an idea of the general qualitative behaviour of $g(r)$. Figure 3.9(a) shows $g(r)$ in the two extreme cases of ideal gas and crystalline solid with lattice spacing a and where r is taken along a lattice axis. Ideally, the pair correlation function of a solid is a series of equally spaced Dirac δ -functions, but thermal fluctuations bring about the observed widening of the peaks. Note that

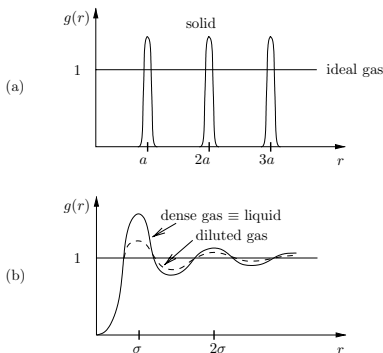


Figure 3.9 Pair correlation function. (a) An ideal gas and a solid, (b) a real gas.

$g(r) = 0$ implies complete anticorrelation, i.e. no particle at a distance r of the reference particle. Figure 3.9(b) shows what we expect for a fluid. We can form a qualitative idea of the correlations with the help of the hard sphere model, where a particle is represented by a hard sphere of diameter σ . This diameter determines the oscillation period of $g(r)$ in the same way that the lattice spacing determines the period for a solid. The average, $g = 1$, means there are no correlations and thus corresponds to the dilute gas case. The amplitude of the oscillations, which decreases as $r \rightarrow \infty$ according to (3.78), increases with the density as the gas moves farther away from the ideal gas case. Beyond the range of the hardcore repulsion, i.e. for $r > \sigma$, the correlation function cannot vanish, contrary to what happens in a solid: we cannot have a fluid full of gaps!

To conclude, we establish a result we shall use later. Consider some function $h(\vec{r}_i, \vec{r}_j)$ and calculate its average value

$$\begin{aligned}
 \left\langle \sum_{i \neq j} h(\vec{r}_i, \vec{r}_j) \right\rangle &= \frac{1}{Z_U} \int_V \frac{d^3 r_1 \dots d^3 r_N}{V^N} \sum_{i \neq j} h(\vec{r}_i, \vec{r}_j) e^{-\beta H_U} \\
 &= \frac{1}{Z_U} \int_V d^3 r d^3 r' \int_V \frac{d^3 r_1 \dots d^3 r_N}{V^N} h(\vec{r}, \vec{r}') \\
 &\quad \times \sum_{i \neq j} \delta(\vec{r} - \vec{r}_i) \delta(\vec{r}' - \vec{r}_j) e^{-\beta H_U} \\
 &= n^2 \int_V d^3 r d^3 r' g(\vec{r}, \vec{r}') h(\vec{r}, \vec{r}') \quad (3.79)
 \end{aligned}$$

3.4.2 Measurement of the pair correlation function

The characterization of spatial correlations in fluids borrows techniques developed in crystallography such as X-ray and neutron scattering. In brief, the measurement is done by illuminating a fluid with a beam of X-rays, which can be considered a plane wave of wave vector $\vec{k}$; the scattered signal is then measured far from the fluid and as a function of θ , the angle with the incident beam (see Figure 3.10). From a fluid atom taken as origin, the detector receives a wave of amplitude $f(\theta)/R$ where $f(\theta)$ is called the scattering amplitude. This quantity is directly related to the differential scattering cross section and contains all the physics of the scattering process. Assuming the collisions to be elastic,³⁶ the angle θ , therefore, selects the scattered wave vectors $\vec{k}'$ such that

$$\vec{k}' - \vec{k} = \vec{q} \quad |\vec{k}'| = |\vec{k}| \quad \vec{q}^2 = 4k^2 \sin^2(\theta/2)$$

³⁶ Inelastic scattering allows the measurement of the dynamic structure factor. See Section 9.1.4.

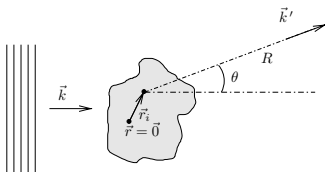


Figure 3.10 X-ray scattering experiment.

The amplitude of a wave scattered by an atom at $\vec{r}_i$ has a phase difference of $(\vec{k}' - \vec{k}) \cdot \vec{r}_i$ relative to the wave scattered by the atom taken as the origin. This means that the scattering amplitude from this atom reads

$$\frac{f(\theta)}{R} e^{i(\vec{k}' - \vec{k}) \cdot \vec{r}_i} = \frac{f(\theta)}{R} e^{i\vec{q} \cdot \vec{r}_i}$$

Since the interaction is the same for all scattering centres, and since we do not know the scatterer, the total scattering amplitude is a sum of amplitudes

$$\mathcal{A}(\theta) = \frac{f(\theta)}{R} \sum_i e^{i\vec{q} \cdot \vec{r}_i} \quad (3.80)$$

A priori, Equation (3.80) contains no information on the statistical fluctuations in the fluid. However, detectors always have an integration time such that the measured intensity is not simply the modulus squared of the (instantaneous) amplitude. Instead, it is given by its (thermodynamic) average value

$$\begin{aligned} I(\theta) &= \langle |\mathcal{A}(\theta)|^2 \rangle = \frac{|f(\theta)|^2}{R^2} \left\langle \sum_{i,j} e^{i\vec{q} \cdot (\vec{r}_i - \vec{r}_j)} \right\rangle \\ &= \frac{|f(\theta)|^2}{R^2} \left(N + \left\langle \sum_{i \neq j} e^{i\vec{q} \cdot (\vec{r}_i - \vec{r}_j)} \right\rangle \right) \end{aligned} \quad (3.81)$$

We now introduce another quantity borrowed from solid state physics, the *structure factor* $S(\vec{q})$

$$I(\theta) = \frac{|f(\theta)|^2}{R^2} N S(\vec{q})$$

Substituting $h(\vec{r}_i, \vec{r}_j) = e^{i\vec{q} \cdot (\vec{r}_i - \vec{r}_j)}$ in (3.79) and using (3.81) we get

$$N S(\vec{q}) = N + n^2 \int_V d^3r d^3r' g(\vec{r}, \vec{r}') e^{i\vec{q} \cdot (\vec{r} - \vec{r}')}$$

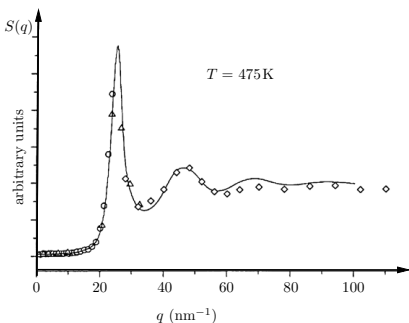


Figure 3.11 The structure factor of liquid lithium measured at the European Synchrotron Radiation Facility (ESRF) at Grenoble, France.

If the fluid is uniform, $g(\vec{r}, \vec{r}')$ is a function only of the separation $(\vec{r} - \vec{r}')$. We can then integrate over all positions of the reference point which yields a factor of V ,

$$S(\vec{q}) = 1 + n \int_V d^3r g(\vec{r}) e^{i\vec{q} \cdot \vec{r}} \quad (3.82)$$

If, in addition, the fluid is isotropic, we have

$$S(q) = 1 + \frac{4\pi n}{q} \int_0^{+\infty} dr r g(r) \sin(qr) \quad (3.83)$$

To within a δ -function, the structure factor (Figure 3.11) is proportional to the Fourier transform of the pair correlation function (review the notion of reciprocal lattice in solid state physics).

3.4.3 Pressure and energy

Since the structure factor is measured experimentally, we know the pair correlation function. It remains for us to express thermodynamic quantities in terms of g . We

begin with the average energy $\langle h(\vec{r}_i, \vec{r}_j) \rangle = U(\vec{r}_i, \vec{r}_j)$ in (3.79))

$$\begin{aligned} E &= \frac{3}{2}NkT + \frac{1}{2} \left\langle \sum_{i \neq j} U(\vec{r}_i - \vec{r}_j) \right\rangle \\ &= \frac{3}{2}NkT + \frac{n^2}{2} \int_V d^3r d^3r' U(\vec{r}, \vec{r}') g(\vec{r}, \vec{r}') \end{aligned} \quad (3.84)$$

For a translation invariant fluid, (3.84) becomes

$$\boxed{\frac{E}{N} = \frac{3}{2}kT + \frac{n}{2} \int_V d^3r U(\vec{r}) g(\vec{r})} \quad (3.85)$$

We now calculate the pressure starting with (3.7)

$$P = \frac{NkT}{V} + kT \left. \frac{\partial}{\partial V} \ln Z_U \right|_T \quad (3.86)$$

We note that Z_U depends on the volume only because of the spatial integration. We take $V = L^3$ and consider the case of a uniform isotropic fluid. Since only relative distances matter, we can rescale $\vec{r}_i$: $\vec{r}_i = L\vec{s}_i$, and so $|\vec{r}_i - \vec{r}_j| = |\vec{s}_i - \vec{s}_j|L$. We then have in the Hamiltonian $H_U = 1/2 \sum U(|\vec{s}_i - \vec{s}_j|L)$, and furthermore

$$\frac{\partial}{\partial V} \rightarrow \frac{L}{3V} \frac{\partial}{\partial L}$$

from which we obtain

$$\begin{aligned} \frac{1}{Z_U} \frac{\partial Z_U}{\partial V} &= \frac{1}{Z_U} \frac{L}{3V} \frac{\partial}{\partial L} \left\{ \int_V \frac{d^3r_1 \dots d^3r_N}{V^N} e^{-\beta H_U} \right\} \\ &= -\frac{1}{Z_U} \frac{L\beta}{6V} \int \prod_{0 \leq |\vec{s}_i| \leq 1} d\vec{s}_i \sum_{i \neq j}^N |\vec{s}_i - \vec{s}_j| U'(|\vec{s}_i - \vec{s}_j|L) e^{-\beta H_U} \\ &= -\frac{1}{Z_U} \frac{\beta}{6V} \int_V \frac{d^3r_1 \dots d^3r_N}{V^N} \sum_{i \neq j} |\vec{r}_i - \vec{r}_j| U'(|\vec{r}_i - \vec{r}_j|) e^{-\beta H_U} \end{aligned}$$

By putting $h(\vec{r}_i, \vec{r}_j) = |\vec{r}_i - \vec{r}_j| U'(|\vec{r}_i - \vec{r}_j|)$ in (3.79), we finally get for the pressure

$$\boxed{P = nkT - \frac{1}{6}n^2 \int_V d^3r g(r) r U'(r)} \quad (3.87)$$

3.5 Chemical potential

3.5.1 Basic formulae

We have already given in Section 1.2.3 the definition (1.11) of the chemical potential μ_i , for particles of type i , at constant energy and volume³⁷

$$\mu_i = -T \left(\frac{\partial S}{\partial N_i} \right)_{E, V, N_j \neq N_i} \quad (3.88)$$

Two other definitions use the free energy F (see (1.24))

$$\mu_i = \left(\frac{\partial F}{\partial N_i} \right)_{T, V, N_j \neq N_i} \quad (3.89)$$

or the Gibbs potential G (see (1.26))

$$\mu_i = \left(\frac{\partial G}{\partial N_i} \right)_{T, P, N_j \neq N_i} \quad (3.90)$$

We emphasize that the variables that are kept constant are different for the different cases. Equation (3.90) yields another demonstration of the Gibbs–Duhem relation (1.41). At constant temperature and pressure, we multiply the number of particles by $(1 + \varepsilon)$ with $\varepsilon \ll 1$: $N_i \rightarrow N_i(1 + \varepsilon)$ or $dN_i = \varepsilon N_i$. Since the Gibbs potential is extensive, this operation gives $G \rightarrow G(1 + \varepsilon)$ and we therefore have

$$dG = \varepsilon G = \sum_i \mu_i (\varepsilon N_i) = \varepsilon \sum_i \mu_i N_i$$

This is the generalization of (1.40) to several particle species³⁸

$$\boxed{G = \sum_i \mu_i N_i} \quad (3.91)$$

By taking the differential of (3.91) and using (1.26) and (3.90) we obtain

$$dG = \sum_i (\mu_i dN_i + N_i d\mu_i) = -S dT + V dP + \sum_i \mu_i dN_i$$

from which we get the Gibbs–Duhem relation

$$\boxed{S dT - V dP + \sum_i N_i d\mu_i = 0} \quad (3.92)$$

³⁷ We do not introduce subsystems here so we can replace the notation $N^{(i)}$ by N_i .

³⁸ The μ_i are not identical to what we would observe for independent subsystems at the same pressure and temperature (see Problem 2.7.10) except in the case of a mixture of ideal gases.

The introduction of the chemical potential allows us to distinguish between two components of the quantity $T dS$: the *conduction* and the *convection* components.³⁹ Consider a system at equilibrium, with a thermostat at temperature T , kept at pressure P and able to exchange particles with a reservoir at chemical potential μ . We take only one kind of particle in order to simplify the discussion. Let $\check{s} = S/N$ and $\check{e} = E/N$ be the entropy and energy per particle. We can write

$$T dS = T d(N\check{s}) = TN d\check{s} + T\check{s} dN \quad (3.93)$$

The first term in (3.93) is the entropy change due to a change in the entropy per particle, in other words this is the conduction term. On the other hand, the second term is the entropy change due to a change in the number of particles, i.e. the convection term. Going back to the equation for dE at constant volume, it is reasonable to regroup all the terms proportional to dN

$$dE = TN d\check{s} + (T\check{s} + \mu) dN = TN d\check{s} + \check{h} dN \quad (3.94)$$

where $\check{h} = \overline{H}/N = \check{e} + Pv$ is the enthalpy per particle. We have used (1.25), the definition of the enthalpy, and (1.40) to write $\mu + T\check{s} = \check{e} + Pv$ with $v = V/N$. The term $\check{h} dN$ has a simple physical interpretation: if dN particles are transported into a given volume by convection, the energy increases by $\check{e} dN$. However, to return to the initial volume, it is necessary to compress the volume by $v dN$, and we need to add into the energy balance the work $Pv dN$ done by the pressure. An example is the Joule–Thomson expansion (Problem 1.7.6). Another example comes from hydrodynamics where one uses in general the Eulerian description in a fixed volume. We then define the energy density $\epsilon = E/V = \check{e}n$ and we write the separation of the conduction and convection terms in the form

$$d\epsilon = Tn d\check{s} + \check{h} dn \quad (3.95)$$

Equation (3.95) explains the rôle played by the enthalpy in fluid mechanics. In a process which conserves entropy, the energy current is given by $(k + h)\vec{u}$ rather than $(k + \epsilon)\vec{u}$ where k is the kinetic energy density, $h = \overline{H}/V$ the enthalpy density and $\vec{u}$ the velocity of the fluid (see Problem 6.5.2).

3.5.2 Coexistence of phases

In this section we are concerned with equilibria between two phases. The chemical potentials μ_1 and μ_2 describe the same kind of particle but in two different phases, e.g. a vapour phase (1) and a liquid phase (2). The equilibrium condition between

³⁹ Consider a fluid in which there is a temperature gradient. If the fluid is kept motionless, thermal equilibrium, which corresponds to a uniform temperature, is reached by conduction. To make the temperature uniform, it is more efficient to mix (stir) the fluid, in other words, to establish convection currents.

the two phases (1.14) means the temperatures are equal and so are the chemical potentials ($\mu_1 = \mu_2$) and consequently the pressures. This equality defines in the (T, P) plane a curve $\mu_1(T, P) = \mu_2(T, P)$ that is the coexistence curve between the two phases. The chemical potentials being equal means, from Equation (3.91), that the Gibbs potentials are equal for the same number of particles in both phases: $G_1(T, P) = G_2(T, P)$. When three phases are in equilibrium, the equality of the chemical potentials

$$\mu_1(T, P) = \mu_2(T, P) = \mu_3(T, P)$$

defines an isolated point in the (T, P) plane called the triple point (Figure 3.8).

We obtain Clapeyron's equation by differentiating along the coexistence curve the relation $\mu_1(T, P) = \mu_2(T, P)$

$$-\check{s}_1 dT + v_1 dP = -\check{s}_2 dT + v_2 dP$$

where $\check{s}_i$ and v_i are the entropy per particle and the specific volume in phase i . This gives

$$\left. \frac{dP}{dT} \right|_{\text{coex}} = \frac{\check{s}_1 - \check{s}_2}{v_1 - v_2} \quad (3.96)$$

The *latent heat* per particle for the transition (2) $\rightarrow$ (1) is $\ell = T(\check{s}_1 - \check{s}_2)$. We can therefore rewrite (3.96) as

$$\left. \frac{dP}{dT} \right|_{\text{coex}} = \frac{\ell}{(v_1 - v_2)T} \quad (3.97)$$

For a liquid (2)–vapour (1) transition, the entropy per particle and the specific volume of the liquid are smaller than those for the vapour, and so $dP/dT > 0$. In the case of a solid (2)–liquid (1) transition, the solid is generally more ordered than the liquid ($\check{s}_2 < \check{s}_1$) and its specific volume smaller ($v_2 < v_1$) which again leads to $dP/dT > 0$. The two exceptions are the water–ice transition where the specific volume of ice is larger than that of water ($v_2 > v_1$), and the case of liquid helium-3 for $T < 0.3$ K where the entropy of the liquid is smaller than that of the solid ($\check{s}_1 < \check{s}_2$) due to the nuclear spins (Problem 5.7.6). In both these cases, $dP/dT < 0$ and for helium-3 we also have $\ell < 0$.

3.5.3 Equilibrium condition at constant pressure

Before proceeding with the study of those transitions, we establish a result that gives the condition for equilibrium of a system $\mathcal{A}$ with a reservoir $\mathcal{R}$ whose temperature T_0 and pressure P_0 are fixed (Figure 3.12). The reservoir is always at

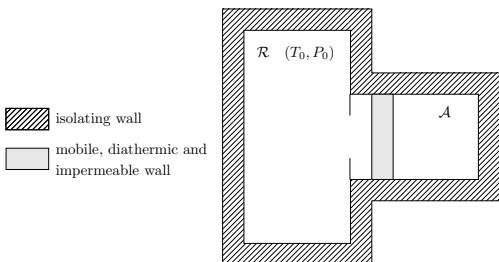


Figure 3.12 Equilibrium at constant temperature and pressure.

equilibrium (its temperature and pressure being well defined and constant) regardless of the arbitrary state of the system $\mathcal{A}$.⁴⁰ The combination $(\mathcal{A} + \mathcal{R})$ is isolated from the outside world. The contact between reservoir and system is assumed to be mobile. With the obvious notation where ΔX denotes the change in a quantity X , which will have a subscript zero if it is a reservoir quantity, and recalling that the entropy of an isolated system can only increase, we have

$$\Delta S_{\text{tot}} = \Delta S + \Delta S_0 \geq 0$$

If $\mathcal{A}$ receives a quantity of heat Q , the reservoir receives $-Q$ and $\Delta S_0 = -Q/T_0$. In the same way, if $\mathcal{A}$ gives the reservoir an amount of work $\mathbb{W} = -P_0\Delta V_0$, then by conservation of total volume we have $\mathbb{W} = P_0\Delta V$ and therefore $Q = \Delta E + P_0\Delta V$. The inequality for the change in the total entropy becomes

$$\Delta S_{\text{tot}} = \Delta S - \frac{Q}{T_0} = \frac{1}{T_0} (T_0 \Delta S - (\Delta E + P_0 \Delta V)) \geq 0 \quad (3.98)$$

By defining the function $G_0 = E - T_0 S + P_0 V$, which reduces to the Gibbs potential when the system and reservoir are in equilibrium ($T = T_0$, $P = P_0$), condition (3.98) becomes

$$\boxed{T_0, P_0 \text{ fixed : } \Delta G_0 \leq 0} \quad (3.99)$$

If the piston is fixed, then with $F_0 = E - T_0 S$ the above condition becomes

$$\boxed{T_0, V \text{ fixed : } \Delta F_0 \leq 0} \quad (3.100)$$

⁴⁰ More precisely, we assume the initial and final states to be in local equilibrium in order for the entropy of the system $\mathcal{A}$ to be well defined. However, as the system is being taken from initial to final states through contact with the reservoir, it can pass through intermediate states that are not in local equilibrium.

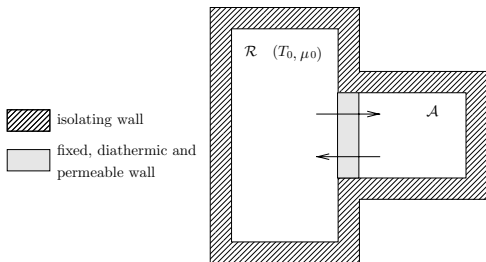


Figure 3.13 Equilibrium at constant temperature and chemical potential.

which we often express by saying that if the temperature and volume are fixed, the system will try to minimize its free energy. The quantity F_0 gives the maximum work that can be supplied by $\mathcal{A}$. In fact, subject to the condition $\Delta S_0 \geq -\Delta S$, the amount of work received by $\mathcal{A}$ is

$$W = \Delta E - Q = \Delta E + T_0 \Delta S_0$$

and it satisfies

$$W \geq \Delta E - T_0 \Delta S = \Delta F_0$$

and so the work furnished is limited by $-\Delta F_0$

$$\mathbb{W} \leq -\Delta F_0 \quad (3.101)$$

This is the same result we found in (1.21).

3.5.4 Equilibrium and stability conditions at constant μ

For the study of phase transitions, it is most convenient to use a variant of the preceding results, by determining the stability condition for a system $\mathcal{A}$ in contact with a reservoir $\mathcal{R}$ whose temperature T_0 and chemical potential μ_0 (and therefore its pressure P_0) are fixed. Indeed, this situation is well suited for the description of phase transitions where particles can go from one phase to the other. Either phase can be considered as the reservoir. The system and the reservoir are put in thermal contact via a porous wall which allows the exchange of particles while the volume of $\mathcal{A}$ stays constant. As above, the entropy of an isolated system can only increase,

and so

$$\Delta S_{\text{tot}} = \Delta S + \Delta S_0 \geq 0$$

Since the reservoir stays at equilibrium we have $\Delta S_0 = Q_0/T_0$ and so

$$T_0 \Delta S_0 = \Delta E_0 - \mu_0 \Delta N_0 = -\Delta E + \mu_0 \Delta N$$

where we have used the conservation of energy and number of particles. We therefore obtain

$$\Delta S_{\text{tot}} = \frac{1}{T_0} (T_0 \Delta S - \Delta E + \mu_0 \Delta N) = -\frac{1}{T_0} \Delta (E - T_0 S - \mu_0 N) \geq 0$$

This result shows that the system is trying to minimize $F_0 - \mu_0 N$

$$\boxed{T_0, \mu_0 \text{ fixed : } \Delta (F_0 - \mu_0 N) \leq 0} \quad (3.102)$$

It is again important to remark that what appear in Equation (3.102) are the temperature and chemical potential of the reservoir. At equilibrium

$$F_0 - \mu_0 N \rightarrow F - \mu N = E - TS - \mu N = -PV \quad (3.103)$$

The quantity $F - \mu N = -PV$ is called the *grand potential*, $\mathcal{J}$, and is a function of (T, V, μ) . Its differential is

$$d\mathcal{J} = -S dT - P dV - N d\mu = -V dP - P dV \quad (3.104)$$

We return now to a system at equilibrium and with free energy F . The free energy is extensive, $F(T, V) = V f(T, n)$, and since the temperature does not play any rôle in what follows we simply write $f(T, n) = f(n)$. From the following identities

$$\left. \frac{\partial n}{\partial N} \right|_V = \frac{1}{V} \quad \left. \frac{\partial n}{\partial V} \right|_N = -\frac{n}{V} \quad (3.105)$$

we obtain

$$\mu = \left. \frac{\partial F}{\partial N} \right|_V = f'(n) \quad (3.106)$$

$$P = - \left. \frac{\partial F}{\partial V} \right|_N = -f + n f'(n) \quad (3.107)$$

These relations can be used for another demonstration of (1.42). From (3.107) we get $(\partial P / \partial n)_T = n f''(n)$, which we use to calculate the coefficient of isothermal expansion, κ_T (1.34)

$$\frac{1}{\kappa_T} = -v \left. \frac{\partial P}{\partial v} \right|_T = n \left. \frac{\partial P}{\partial n} \right|_T = n^2 f''(n) = n^2 \left. \frac{\partial \mu}{\partial n} \right|_T \quad (3.108)$$

This equation shows that f is a convex function of n ($f''(n) \geq 0$) because a condition for stability is $\kappa_T \geq 0$: when $\partial P / \partial n \propto f''(n)$ becomes negative, this signals a phase transition. The pressure P is an increasing function of n on both the high density side (e.g. liquid phase) and the low density side (e.g. gaseous phase). The pressure becomes a decreasing function in the interval $\bar{n}_1 \leq n \leq \bar{n}_2$, which represents an instability zone. The densities $\bar{n}_1$ and $\bar{n}_2$ are inflexion points of the function f (Figure 3.14) called *spinodal points*. If the chemical potential μ_0 is imposed by a reservoir, then since, on the one hand, $(f - \mu_0 n)$ must be the same in both phases⁴¹ and, on the other hand, we minimize $(f - \mu_0 n)$ according to (3.102), we conclude that $f' = \mu_0$. These two conditions allow us to construct the common tangent (see Figure 3.14(b)), which determines the densities n_1 and n_2 in phases (1) and (2)

$$\frac{f_2 - f_1}{n_2 - n_1} = \mu_0 \quad (3.109)$$

These same densities can also be obtained from the Maxwell construction (Figure 3.14(c))

$$\int_{n_1}^{n_2} dn (\mu - \mu_0) = \int_{n_1}^{n_2} dn (f'(n) - \mu_0) = [f - \mu_0 n]_1^2 = 0 \quad (3.110)$$

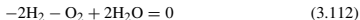
This equation means that the shaded areas in Figure 3.14(c) are equal. The zones defined by $n_1 \leq n \leq \bar{n}_1$ and $\bar{n}_2 \leq n \leq n_2$ are zones of metastability.

3.5.5 Chemical reactions

The chemical potential is an essential tool for studying chemical reactions. We limit ourselves to a few simple examples by considering only reactions in the gaseous phase. Let $B_1, B_2, \dots, B_m$ be the m types of molecules involved in the reaction, for example $B_1 = \text{H}_2$, $B_2 = \text{O}_2$ and $B_3 = \text{H}_2\text{O}$ for the reaction



It is convenient to rewrite (3.111) in the symbolic form



⁴¹ Let x be the fraction (in mass) of molecules in phase 1, $(1 - x)$ the fraction in phase 2, $f - \mu_0 n = x[f_1 - \mu_0 n_1] + (1 - x)[f_2 - \mu_0 n_2]$ should be stationary in the variable x .

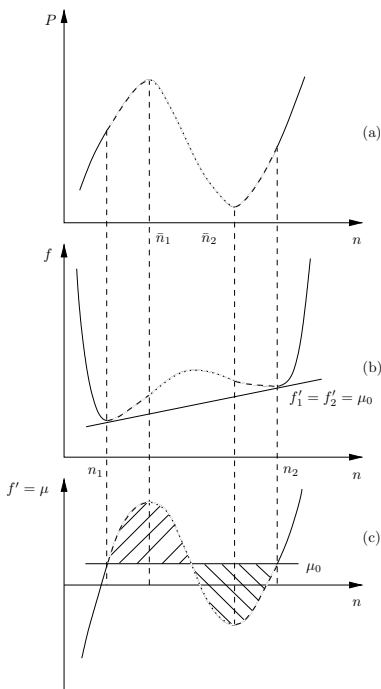


Figure 3.14 (a) The stability curve $P(n)$: the dotted line shows the unstable region, the dashed line the metastable one. The spinodal points $\bar{n}_1$ and $\bar{n}_2$ correspond to the extrema of P (or f'). (b) Determination of the densities n_1 and n_2 of the phases 1 and 2 using the common tangent construction. The spinodal points are the inflexion points of f . (c) Determination of n_1 and n_2 using the Maxwell construction.

and in the same way we represent a general chemical reaction by

$$\sum_{i=1}^m b_i B_i = 0 \quad (3.113)$$

The integers b_i are the stoichiometric coefficients that are needed to balance the reaction. In example (3.112) we have $b_1 = -2$, $b_2 = -1$, $b_3 = 2$. For an isolated system ($dE = dV = 0$) the equilibrium condition is that of maximum entropy, $dS = 0$, which yields $\sum_{i=1}^m \mu_i dN_i = 0$, where N_i is the number of molecules of type B_i . In order to ensure that the reaction be balanced, the variations dN_i must be proportional to the coefficients b_i : $dN_i = \varepsilon b_i$. This leads to the important result

$$\boxed{\sum_{i=1}^m b_i \mu_i = 0} \quad (3.114)$$

In this demonstration, the chemical potentials are functions of E and V , the natural variables for the entropy. We can also obtain Equation (3.114) by writing $dF = 0$ at constant temperature and volume, or $dG = 0$ at constant temperature and pressure. This would give the chemical potentials as functions of (T, V) and (T, P) respectively. Equation (3.114) is very general, it is valid for reactions in gases, dense or dilute, and for reactions taking place in solution.

The *law of mass action*⁴² is obtained by considering the reaction to take place in the gaseous phase, and by assuming that we can ignore the interactions between molecules and the effects of quantum statistics. In other words, consider the approximation of the classical ideal gas. Under these conditions, the partition function for the m species is

$$Z = \prod_{i=1}^m \frac{(V \zeta_i)^{N_i}}{N_i!} \quad (3.115)$$

where, from (3.13)

$$\zeta_i = \left(\frac{1}{\lambda_i^3} \right) \sum_s \exp \left(-\beta \varepsilon_s^{\text{int}(i)} \right) \quad (3.116)$$

In this equation, λ_i is the thermal wavelength (3.11) of molecules of type i , the sum over s runs over all internal molecular degrees of freedom such as vibration and rotation, and $\varepsilon_s^{\text{int}(i)}$ are the energies of the corresponding levels. The chemical

⁴² Yet another mysterious terminology!

potentials μ_i are then given by

$$\mu_i = \left. \frac{\partial F}{\partial N_i} \right|_{V, T, N_j \neq N_i} = -kT \ln \frac{V \zeta_i}{N_i} \quad (3.117)$$

This gives the equilibrium condition

$$\sum_{i=1}^N b_i \ln \frac{\zeta_i}{n_i} = 0$$

which, upon exponentiation, becomes

$$\boxed{n_1^{b_1} \dots n_m^{b_m} = \zeta_1^{b_1} \dots \zeta_m^{b_m} = K(T)} \quad (3.118)$$

since the ζ_i are functions only of T . Equation (3.118) expresses the law of mass action for the concentrations $n_1, \dots, n_m$: the product of concentrations n_i each raised to the power b_i is equal to a function $K(T)$ which depends only on the temperature.

As an application, we demonstrate the Saha law that gives the dissociation percentage of atomic hydrogen H into free electrons e and protons p ($e + p \rightleftharpoons H$) as a function of the temperature. The previously defined quantities ζ are given as follows for the electrons, protons and atomic hydrogen

$$\zeta_e = \frac{2}{\lambda_e^3} = 2 \left(\frac{2\pi m_e}{\beta h^2} \right)^{3/2} \quad (3.119)$$

$$\zeta_p = \frac{2}{\lambda_p^3} = 2 \left(\frac{2\pi m_p}{\beta h^2} \right)^{3/2} \quad (3.120)$$

$$\zeta_H = 4 \left(\frac{2\pi m_H}{\beta h^2} \right)^{3/2} \sum_n \exp(-\beta E_n) \quad (3.121)$$

The factors 2 for the electron and proton and 4 for the hydrogen come from the spin degeneracies. The sum over n in (3.121) runs over the hydrogen energy levels in accordance with (3.116). The ground state of the hydrogen atom has $E_0 = -13.6$ eV, which corresponds to a temperature of 1.6×10^5 K. If we assume $T \ll 10^5$ K, the excited levels can be ignored because their weights are negligible in (3.121). In addition we can take the masses of the proton and atomic hydrogen to be the same, $m_p \simeq m_H$, which gives

$$\zeta_H \simeq 4 \left(\frac{2\pi m_p}{\beta h^2} \right)^{3/2} \exp(\beta |E_0|)$$

Using (3.118), we obtain the Saha law

$$\frac{n_e n_p}{n_H} \simeq \left(\frac{2\pi m_e}{\beta h^2} \right)^{3/2} \exp(-\beta |E_0|) \quad (3.122)$$

If the only source of electrons and protons is the dissociation of neutral hydrogen, then $n_e = n_p$ and (3.122) becomes

$$n_e = \left(\frac{2\pi m_e}{\beta h^2} \right)^{3/4} \exp(-\beta |E_0|/2) n_H^{1/2} \quad (3.123)$$

This equation illustrates the energy–entropy competition involved in the minimization of the free energy $E - TS$. At low temperature, it is advantageous to minimize E thus favouring the formation of atomic hydrogen; at high temperature, it is more advantageous to increase entropy thus dissociating the atoms.

3.6 Grand canonical ensemble

3.6.1 Grand partition function

Following the general framework outlined in Chapter 2, we now examine in more detail the grand canonical ensemble. In other words, we study a system that can exchange both energy and particles with a reservoir. Therefore, this fixes the average values $E = \langle H \rangle$ and $\bar{N} = \langle N \rangle$ of the energy and particle number.⁴³ We now have to change notation from that used in Chapter 2. Since N designates a well-defined integer number of particles, the eigenvalue of $\mathbf{N}$, it is necessary to use a different notation, $\bar{N}$, for the average value. Following the construction of the Boltzmann distribution in Chapter 2, the density operator in the grand canonical ensemble is given in terms of the two Lagrange multipliers, α and β , by

$$D = \frac{1}{Q} \exp(-\beta H + \alpha \mathbf{N}) \quad Q = \text{Tr} \exp(-\beta H + \alpha \mathbf{N}) \quad (3.124)$$

Q is called the *grand partition function*. The density operator D acts in a Hilbert space $\mathcal{H}$ that is the direct sum⁴⁴ of Hilbert spaces $\mathcal{H}^{(N)}$ each of which describes a

⁴³ The correct mathematical description of the grand canonical ensemble requires, in principle, a formalism which allows the number of particles to change, which is not the case for elementary quantum mechanics. Nonetheless, such a formalism exists and is (improperly) called ‘second quantization’. We will construct this for a simple case in Section 5.4.1. We do not need this formalism here, we only need to know that it exists.

⁴⁴ One should not confuse the direct sum of spaces with their tensor product. For example, for two spaces of dimensions N_1 and N_2 , the dimension of the direct sum space is $N_1 + N_2$, whereas the dimension of the tensor product space is $N_1 N_2$. The trace of a direct sum operator is $\text{Tr} \mathbf{A} = \text{Tr} \mathbf{A}^{(1)} + \text{Tr} \mathbf{A}^{(2)}$ while that for a tensor product operator is $\text{Tr} \mathbf{A} = \text{Tr} \mathbf{A}^{(1)} \text{Tr} \mathbf{A}^{(2)}$.

fixed number, N , of particles

$$\mathcal{H} = \bigoplus_N \mathcal{H}^{(N)}$$

In the ‘second quantization’ formalism, this space is called a Fock space.

In non-relativistic quantum mechanics, the number of particles is conserved: $[H, N] = 0$, the Hamiltonian commutes with the number operator, which is not the case in relativistic quantum mechanics. If we consider, for example, electrons and positrons, we find that the number of these particles is not conserved individually; what is conserved is the difference in the numbers of electrons and positrons due to conservation of electric charge. In the Boltzmann distribution, the Lagrange multiplier α (or the directly related chemical potential μ) would then multiply the difference, $(N_- - N_+)$, between the number operators for the electrons and positrons. Generally, the chemical potential is associated with a conservation law. In elementary quantum mechanics, to which we will now limit ourselves, this is the conservation of the number of particles.

We now write down the explicit form of the grand canonical partition function. Since $[H, N] = 0$, we can diagonalize simultaneously these two operators. Let $|N, r\rangle$ be a basis labeled by the eigenvalue, N , of the number operator, N , and r which labels the energies E_r , which are functions of N and external parameters x_i

$$\begin{aligned} H|N, r\rangle &= E_r(N, x_i)|N, r\rangle \\ N|N, r\rangle &= N|N, r\rangle \end{aligned} \quad (3.125)$$

In the N -particle Hilbert space $\mathcal{H}^{(N)}$ we have

$$\langle N, r | e^{-\beta H + \alpha N} | N, r \rangle = e^{\alpha N} e^{-\beta E_r(N, x_i)}$$

which gives the following trace in $\mathcal{H}^{(N)}$

$$\text{Tr}_N e^{-\beta H + \alpha N} = e^{\alpha N} \sum_r e^{-\beta E_r(N, x_i)}$$

The grand partition function, which is a function of α , β and x_i , is obtained by summing over N

$$\mathcal{Q}(\alpha, \beta, x_i) = \sum_{N=0}^{\infty} e^{\alpha N} \sum_r e^{-\beta E_r(N, x_i)} \quad (3.126)$$

The sum over r in (3.126) is simply the canonical partition function. We can then write (3.126) in the form

$$\mathcal{Q}(\alpha, \beta, x_i) = \sum_{N=0}^{\infty} e^{\alpha N} Z_N(\beta, x_i) \quad (3.127)$$

By proceeding at fixed external parameters, we have expressed in Chapter 2 Equation (2.97) the Lagrange multipliers, α and β , in terms of the temperature and chemical potential

$$\beta = \frac{1}{kT} \quad \alpha = \frac{\mu}{kT} \quad (3.128)$$

From Equations (2.66) and (2.95), we obtain the average values of the energy, the particle number and the conjugate forces X_i

$$E = - \left. \frac{\partial \ln \mathcal{Q}}{\partial \beta} \right|_{\alpha, x_i} \quad \bar{N} = \left. \frac{\partial \ln \mathcal{Q}}{\partial \alpha} \right|_{\beta, x_i} \quad X_i = - \left. \frac{1}{\beta} \frac{\partial \ln \mathcal{Q}}{\partial x_i} \right|_{\alpha, \beta, x_{j \neq i}} \quad (3.129)$$

In particular, for $x_i = -V$ (to which we will limit ourselves in what follows) the pressure is given by

$$P = \left. \frac{1}{\beta} \frac{\partial \ln \mathcal{Q}}{\partial V} \right|_{\alpha, \beta} \quad (3.130)$$

The entropy is the Legendre transform of the partition function (cf. (2.67))

$$\frac{1}{k} S = \ln \mathcal{Q} + \beta E - \alpha \bar{N} \quad (3.131)$$

which gives the entropy differential

$$\begin{aligned} \frac{1}{k} dS &= d \ln \mathcal{Q} + d(\beta E) - d(\alpha \bar{N}) \\ &= \beta dE + \beta P dV - \alpha d\bar{N} \end{aligned} \quad (3.132)$$

This is nothing but Equation (1.18)

$$T dS = dE + P dV - \mu d\bar{N}$$

Finally, we remark that (3.131) gives a relation between the grand potential, $\mathcal{J}$, and the logarithm of the grand partition function. Using μ and T instead of α and β , we get

$$\boxed{\mathcal{J}(T, V, \mu) = -kT \ln \mathcal{Q} = -PV} \quad (3.133)$$

This equation is the counterpart of Equation (3.6) for the canonical ensemble. The grand potential, $\mathcal{J}$, plays in the grand canonical ensemble the same rôle played

by the free energy, F , in the canonical ensemble. Often, instead of the chemical potential, we use the *fugacity* z

$$z = e^{\beta\mu} = e^{\alpha} \quad (3.134)$$

which permits us to write (3.127) in the form that we will use preferentially in what follows

$$\mathcal{Q}(z, \beta, V) = \sum_{N=0}^{\infty} z^N Z_N(\beta, V) \quad (3.135)$$

The average values then become⁴⁵

$$E = - \left. \frac{\partial \ln \mathcal{Q}}{\partial \beta} \right|_{z, V} \quad \bar{N} = z \left. \frac{\partial \ln \mathcal{Q}}{\partial z} \right|_{\beta, V} \quad P = \frac{1}{\beta} \left. \frac{\partial \ln \mathcal{Q}}{\partial V} \right|_{z, \beta} \quad (3.136)$$

3.6.2 Mono-atomic ideal gas

As a simple application, we calculate the grand partition function for an ideal mono-atomic gas. Using Equations (3.13) and (3.135) we obtain

$$\mathcal{Q} = \sum_{N=0}^{\infty} \frac{z^N}{N!} \left(\frac{V}{\lambda^3} \right)^N = \exp \left(\frac{zV}{\lambda^3} \right) \quad (3.137)$$

With Equations (3.133) and (3.136) this leads to

$$\bar{N} = z \frac{\partial}{\partial z} \left(\frac{zV}{\lambda^3} \right) = \frac{zV}{\lambda^3} \quad \text{and} \quad \left(\frac{zV}{\lambda^3} \right) = \beta PV \quad (3.138)$$

which is the ideal gas law

$$PV = \bar{N}kT$$

This simple example illustrates the strategy to be followed in the grand canonical ensemble. With μ or z given, we calculate the average value $\bar{N}$ using (3.136), which then gives μ or z in terms of $\bar{N}$. This value is then put into Equation (3.133) of the grand potential, which gives the equation of state, i.e. PV as a function of T , V and $\bar{N}$.

⁴⁵ Care should be taken because a derivative at fixed μ is not the same as at fixed z or α : $(\partial \ln \mathcal{Q} / \partial \beta)_{\mu, V} = -E + \mu \bar{N}$.

In the ideal gas case, the relation between the density and fugacity is particularly simple. From (3.138) we get

$$z = \beta P \lambda^3 = n \lambda^3 \sim \left(\frac{\lambda}{d}\right)^3 \quad (3.139)$$

where $n = \bar{N}/V$ is the density and $d \sim n^{-1/3}$ the average distance between two molecules. Of course, Equations (3.139) and (3.18) are identical, which means that the classical ideal gas approximation corresponds to the condition $z \ll 1$. Finally, we note that the distribution function, $f(\vec{r}, \vec{p})$, normalized to

$$\int d^3p f(\vec{r}, \vec{p}) = n \quad (3.140)$$

is given by

$$f(\vec{r}, \vec{p}) = \frac{z}{h^3} \exp\left(-\frac{\beta p^2}{2m}\right) = \frac{1}{h^3} \exp\left(\alpha - \frac{\beta p^2}{2m}\right) \quad (3.141)$$

This relation will be useful in Chapter 8.

3.6.3 Thermodynamics and fluctuations

It remains to show that thermodynamics in the grand canonical ensemble is identical to that in the canonical ensemble for large systems. Physically, this equivalence comes from the fact that fluctuations in the number of particles, $\Delta N = \langle (N - \bar{N})^2 \rangle^{1/2}$, are of the order of $\sqrt{\bar{N}}$, which is negligible in the limit $\bar{N} \rightarrow \infty$: $\Delta N / \bar{N} \rightarrow 0$. Fixing the value N is equivalent to fixing the average value $\bar{N}$ when the variance is negligible. In order to give a precise argument, we study the distribution $P(N)$ of the variable N . According to (3.127), we have

$$P(N) \propto e^{\beta \mu N} e^{-\beta F_N} = P'(N)$$

where $F_N = -kT \ln Z_N$ is the free energy in the canonical ensemble. It is convenient to use the logarithm of $P'(N)$ rather than $P'(N)$ itself. Let $\tilde{N}$ be the most probable value of N that then satisfies

$$\left. \frac{\partial \ln P'(N)}{\partial N} \right|_{N=\tilde{N}} = \beta \mu - \beta \left. \frac{\partial F_N}{\partial N} \right|_{N=\tilde{N}} = \beta (\mu - \mu_{\text{can}}(\tilde{N})) = 0$$

Therefore, $\tilde{N}$ is such that the chemical potential μ of the reservoir is the same as $\mu_{\text{can}}(\tilde{N})$, the chemical potential in the canonical ensemble. Using (3.108) we get the second derivative of $\ln P'(N)$

$$\frac{\partial^2 \ln P'(N)}{\partial N^2} = -\beta \frac{\partial^2 F_N}{\partial N^2} = -\frac{\beta}{V} f''(n) = -\frac{\beta v}{\tilde{N} \kappa_T}$$

where $v = V/\tilde{N}$ is the specific volume and κ_T the isothermal compressibility (1.34). Writing the Taylor series of $\ln P'(N)$ in the neighbourhood of $N = \tilde{N}$, we have

$$\ln P'(N) \simeq \ln P'(\tilde{N}) - \frac{\beta v}{2\tilde{N}\kappa_T}(N - \tilde{N})^2 \quad (3.142)$$

This equation shows that $P'(N)$ is approximately a Gaussian centred at $N = \tilde{N}$ and whose variance is

$$(\Delta N)^2 = \frac{\kappa_T \bar{N}}{\beta v} = \frac{\bar{N}}{\beta v^2 (-\partial P / \partial v)_T} \quad (3.143)$$

Since $\Delta N / \bar{N} \rightarrow 0$, we can take the average value to be equal to the most probable value: $\bar{N} = \tilde{N}$. For the same reason, the numerical value of the Gaussian approximation of $P'(N)$ is negligible for $N \leq 0$, and so we can replace the fluctuation interval of N , $[0, +\infty]$, by the expanded interval $[-\infty, +\infty]$. If κ_T is finite, i.e. if $(\partial P / \partial v)_T \neq 0$, the particle number distribution is very sharply peaked around its average value $\bar{N}$. With this assumption for κ_T , we can calculate $\mathcal{Q}$ using (3.127) by replacing the sum over N by an integral

$$\begin{aligned} \mathcal{Q} &= \sum_{N=0}^{\infty} P'(N) \simeq P'(\bar{N}) \int_{-\infty}^{\infty} dN \exp \left[-\frac{(N - \bar{N})^2}{2(\Delta N)^2} \right] \\ &= \left(\frac{2\pi \bar{N} \kappa_T}{\beta v} \right)^{1/2} P'(\bar{N}) \end{aligned} \quad (3.144)$$

In the calculation of $\ln \mathcal{Q}$, the first term of (3.144) is of the order of $\ln \bar{N}$ and is therefore negligible compared to the second term which is of the order $\bar{N}$

$$\begin{aligned} \ln \mathcal{Q} &= \beta P V|_{\text{g.can}} \simeq \ln P'(\bar{N}) = \beta(\mu \bar{N} - F_{\bar{N}}) \\ &= \beta(\mu_{\text{can}} \bar{N} - F_{\bar{N}}) = \beta(G - F) = \beta P V|_{\text{can}} \end{aligned} \quad (3.145)$$

As long as $(\partial P / \partial v)_T \neq 0$ and $\bar{N} \rightarrow \infty$, the canonical and grand canonical ensembles give the same value for PV , and thus the same value for the grand potential and the same thermodynamics. A somewhat longer analysis leads to the same conclusion when $(\partial P / \partial v)_T = 0$, i.e. for a phase transition. Close to a critical point, where $(\partial P / \partial v)_T \simeq 0$, density fluctuations become very large leading to the phenomenon of critical opalescence.

The preceding calculation and Equation (3.10) lead to the conclusion that for homogeneous substances, the stability conditions (1.52) follow from statistical

mechanics

$$C_V \propto \langle (H - E)^2 \rangle \geq 0 \quad \kappa_T \propto \langle (N - \bar{N})^2 \rangle \geq 0 \quad (3.146)$$

The specific heat is related to the energy fluctuations and the isothermal compressibility to the particle number fluctuations.

3.7 Exercises

3.7.1 Density of states

Consider a dilute gas in contact with a heat reservoir at temperature T ($\beta = 1/kT$) and let $\mathcal{P}(\varepsilon) d\varepsilon$ be the probability that a molecule has an energy between ε and $\varepsilon + d\varepsilon$. Why is it not correct to write

$$\mathcal{P}(\varepsilon) d\varepsilon \propto e^{-\beta\varepsilon} d\varepsilon \quad (3.147)$$

3.7.2 Equation of state for the Einstein model of a solid

A crystalline solid contains N atoms that can vibrate around their equilibrium lattice sites. As a first approximation, we may view the solid as $3N$ independent, identical but distinguishable harmonic oscillators. In the Einstein model, all these oscillators have the same frequency ω .

1. Calculate the partition function $\zeta(\beta)$ for a harmonic oscillator. Use ζ to calculate the partition function Z_N of the solid. Derive the expression for the internal energy E and verify that the specific heat has the form (3.68).
2. We introduce the Grüneisen constant γ by assuming that the frequency ω is a power law of the volume V of the solid,

$$\frac{\partial \ln \omega}{\partial \ln V} \simeq -\gamma$$

Show that the pressure is given by

$$P = \gamma \frac{E}{V}$$

and that

$$\gamma = \frac{\alpha V}{\kappa_T C_V} \quad (3.148)$$

where α and κ_T are the thermodynamic coefficients defined in (1.33) and (1.34).

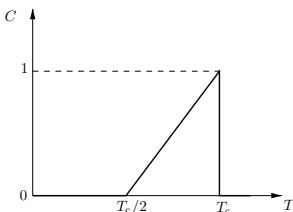


Figure 3.15 Approximate behaviour of the specific heat of a ferromagnet as a function of temperature.

3.7.3 Specific heat of a ferromagnetic crystal

A rough approximation of the specific heat C due to atomic spins in a ferromagnetic crystal is (Figure 3.15)

$$C = \begin{cases} NkA \left(2\frac{T}{T_c} - 1 \right) & \text{for } T_c/2 \leq T \leq T_c \\ 0 & \text{elsewhere} \end{cases}$$

T_c is the Curie temperature, N the number of atoms and A a constant. The atoms of the system are modeled by N Ising spins placed on the lattice sites of the crystal. Estimate A .

3.7.4 Nuclear specific heat of a metal

Consider a metal whose atoms have nuclear spin 1 which can be in one of three quantum states: $m = -1$, $m = 0$ or $m = +1$. Each atom also carries an electric quadrupole moment that interacts with the internal electric field of the crystal. We ignore spin-spin interactions. Because of the interactions between the electric field and the quadrupoles, the states $m = \pm 1$ have the same energy, higher by an amount ε than the energy of the $m = 0$ state which we take as the reference state.

1. For a mole of these atoms, calculate the average energy E and the specific heat C due to the nuclear quadrupoles.

2. How does the specific heat behave at high temperature? At low temperature? Draw schematically C as a function of $\beta\varepsilon$. Evaluate the integral

$$\int_0^{\infty} dT \frac{C(T)}{T}$$

3. The gallium atom has an electric quadrupole moment, and we can measure its specific heat. For a temperature range from 3 to 20 mK, we have experimentally⁴⁶

$$C_{\text{exp}}(T) = 4.3 \times 10^{-4} T^{-2} + 0.596 T$$

where $C_{\text{exp}}(T)$ is expressed in mJ K/mol^{-1} . Interpret the behaviour of $C_{\text{exp}}(T)$. Estimate the value of ε/k in K and of ε in eV.

3.7.5 Solid and liquid vapour pressures

Consider a container of volume V in which are placed N atoms. These atoms are in two coexisting phases: solid and gas. The pressure P and temperature T of the gas can be perfectly controlled, and the volume occupied by the solid is very small compared with that of the gas. We call u_0 the ionization energy per atom ($u_0 > 0$) of the solid: this is the energy needed to remove an atom from the solid and take it to infinity at zero temperature. Energy and atoms may be exchanged between gas and solid thus allowing equilibrium to be established. We shall calculate the saturated vapour pressure of the solid. We recall that the chemical potential $\mu_{\text{IG}}(T, V)$ of a mono-atomic ideal gas with no internal degrees of freedom is given by (3.18).

1. Let μ_s and μ_g be the chemical potentials of this substance in its solid and gaseous phases respectively. What precaution should be taken to make sure the two chemical potentials are equal at equilibrium?
2. We assume the solid is incompressible with specific volume v_s and specific heat per atom $c(T)$. Calculate the Gibbs potential of the solid and show that its chemical potential may be written as

$$\mu_s(T) = v_s P - u_0 - \int_0^T dT' \left(\frac{T}{T'} - 1 \right) c(T') \quad (3.149)$$

Write an implicit equation for the vapour pressure $P(T)$ of the solid at equilibrium with its gaseous phase. Give the explicit expression for $P(T)$ in the approximation where v_s is negligible compared with the specific volume of the gas.

⁴⁶ The experimental conditions are such that the superconductivity normally present for gallium at such low temperatures is suppressed: gallium behaves as a normal metal.

3. We now establish the form of $P(T)$ when we have equilibrium between a gas and a liquid. To this end we use a simple microscopic model for the liquid phase while keeping the mono-atomic ideal gas approximation for the gas. Each atom of the liquid phase occupies a volume v_0 , the liquid is incompressible and has a total volume Nv_0 for N atoms. The atoms are independent each with a potential energy $-u_0$, $u_0 > 0$. Clearly, this is a very rough model for a liquid. Calculate the partition function of the liquid. Deduce the expressions for the Gibbs potential G_1 and the chemical potential μ_1 of the liquid as a function of pressure P

$$\mu_1 = Pv_0 - u_0 - kT \ln \left(\frac{ev_0}{\lambda^3} \right)$$

Why is it not correct to use $\mu_1 = \partial F_N / \partial N|_V$, where F_N is the free energy? Applying the condition that the chemical potentials of the liquid and gaseous phases are equal, derive an implicit equation for the vapour pressure $P(T)$ at temperature T . Assuming that $Pv_0/(kT) \ll 1$, show that we have the following explicit expression

$$P(T) = \frac{kT}{ev_0} e^{-u_0/(kT)} \quad (3.150)$$

Is the condition $Pv_0/(kT) \ll 1$ reasonable physically? For the remainder of the problem we shall assume this condition to hold.

4. Calculate the entropy per atom for the gas (σ_g) and for the liquid (σ_l). Find the expression for the latent heat of vaporization per atom, ℓ . Is the Clapeyron relation

$$\frac{dP}{dT} = \frac{\ell}{T(v_g - v_l)} = \frac{\ell}{T(v - v_0)}$$

satisfied? Hint: (i) $v_0 \ll v$ and (ii) show that

$$\ln \frac{v}{v_0} = 1 + \frac{u_0}{kT}$$

In the case of water boiling at atmospheric pressure (10^5 Pa), calculate $\mu_0/(kT)$

- (i) from the latent heat of vaporization of 40.7 kJ/mole,
- (ii) from the equation of the vaporization curve (3.150).

Compare the two results.

3.7.6 Electron trapping in a solid

Consider a solid with A identical and independent sites each of which is capable of trapping at most one electron. We use for the magnetic moment of the electrons an Ising spin model: the electron spin magnetic moment can only take the two

values $\vec{\mu} = \pm \tilde{\mu} \hat{z}$ corresponding to the two states ($\pm$). In the presence of an external magnetic field $\vec{B} = B \hat{z}$, the energy of a trapped electron depends on the orientation of the magnetic moment

$$\varepsilon_{\pm} = -u_0 \mp \tilde{\mu} B$$

1. Consider first the case where the number of trapped electrons is fixed at N ($N \leq A$). Calculate $Z_N(T, A)$, the partition function of the system of N trapped electrons. Calculate E , the average energy of the system, and $\bar{N}_+$, the average number of electrons with a magnetic moment in the state (+).

2. Calculate the chemical potential μ of the trapped electrons. Give the expression for N in terms of μ .

3. We now consider the grand canonical case. We assume the solid is in equilibrium with a gas of electrons that we consider to be ideal and classical. This imposes a chemical potential μ . Calculate the grand partition function of the system $\mathcal{Q}(T, A, \mu)$

(i) by using the partition function Z_N explicitly,

(ii) by first calculating the grand partition function of a single trap $\xi(T, A, \mu)$.

4. Use ξ to calculate the probabilities p_+ and p_- for a trap to be occupied by an electron with a magnetic moment parallel and antiparallel to $\vec{B}$. Find the expressions for $\bar{N}_+$, $\bar{N}_-$ and $\bar{N}$. Verify that the expressions for E obtained in the canonical and grand canonical ensembles are identical if we take $N = \bar{N}$.

3.8 Problems

3.8.1 One-dimensional Ising model

A one-dimensional lattice consists of $N + 1$ equally spaced Ising spins (Figure 3.16) coupled by nearest-neighbour exchange interactions. The Hamiltonian of the system is

$$H = -J \sum_{j=1}^N S_j S_{j+1}$$

where J is a positive constant.



Figure 3.16 One-dimensional lattice of Ising spins.

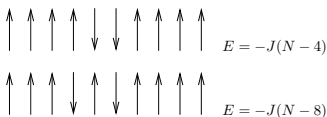


Figure 3.17 Two configurations with the same number of flipped spins but with different energies.

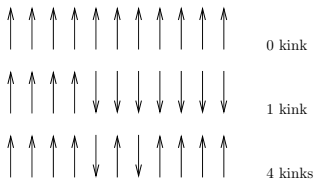


Figure 3.18 Number of kinks in three different configurations.

A Preliminaries

We can see from Figure 3.17 that, in this problem, flipping two neighbouring spins and flipping two distant spins generates configurations with different energies. The energy of a configuration is determined by the number of nearest-neighbour pairs of antiparallel spins, called ‘kinks’. To obtain an elementary kink, one flips all spins to the left (or right) of a chosen site (Figure 3.18). A configuration with m kinks has the energy

$$E_m = -NJ + m2J$$

Represent schematically the configurations corresponding respectively to the states of minimum and maximum energy $E_{\min}$ and $E_{\max}$. Give the number of kinks and the degeneracies associated with these two states.

B Fixed number of kinks

We fix the number of kinks to $m = n$ and, therefore, the energy is fixed at $E = E_n$.

1. Calculate Ω_n , the number of states with n kinks and find the expression for the corresponding entropy $S(n, N)$ assuming $n, N \gg 1$.

2. Calculate the equilibrium temperature $\beta = 1/kT$ as a function of n and N . Verify that it can also be written as⁴⁷

$$\beta = -\frac{1}{J} \tanh^{-1} \left(\frac{E}{NJ} \right) \quad (3.151)$$

It seems natural to associate high temperature with high energy. Is this confirmed for this spin lattice? If no, why?

3. Calculate the probability p that two neighbouring spins are antiparallel.

C Arbitrary number of kinks

We now consider the problem in the canonical ensemble. The temperature is fixed but the energy is known only on average.

1. Compare the changes in the entropy and energy when the system goes from no kinks to one kink. Verify that, for $T = 0$, there is no spontaneous magnetization in this one-dimensional system (cf. Section 3.1.4).
2. Show that the partition function of the one-dimensional Ising model may be written as

$$Z = 2^{(N+1)} (\cosh \beta J)^N$$

Can one make an analogy with paramagnetism and explain why the solution of the one-dimensional Ising model is trivial?

3. Calculate the average energy E and the average number of kinks $\langle m \rangle$. Find an expression for the probability p defined above. Sketch and comment on the curve representing p as a function of T . Show that the maximal entropy is obtained in the high temperature limit.
4. How can we justify the statement that the points of view adopted in parts B and C are equivalent?

3.8.2 Negative temperatures

For some physical systems, during transient periods that can last several minutes, the nuclear spins and the crystalline lattice can separately reach independent states of thermodynamic equilibrium.⁴⁸ The necessary condition for this to happen is that

⁴⁷ We give the following relation: $\tanh^{-1} x = \frac{1}{2} \ln \frac{1+x}{1-x}$. This function appears systematically in all problems concerning magnetism in spin-1/2 systems. See also Problem 3.8.2.

⁴⁸ Contrary to what is often stated, this may happen without the need for each subsystem to pass many times through all accessible states. Such a condition would require a relaxation time several times the age of the universe since the number of states for the spin system is of the order of $2^{10^{23}}$.

the spin-lattice relaxation time τ be long compared to the separate spin and lattice relaxation times τ_s and τ_l respectively. If the system is in a state of global equilibrium, the sudden change of an external parameter (for example a magnetic field) is followed by a transient period during which the two subsystems are isolated one from the other and may thus attain independent equilibrium states characterized by different temperatures, even *negative* temperature for the spin lattice. In this problem we illustrate this *a priori* surprising property by using the Ising spin model of a paramagnetic crystal (Sections 3.1.3 and 3.1.4)

$$H = -\mu B \sum_{i=1}^N S_i$$

1. Calculate the partition function, $\zeta(\beta, B)$, of a single spin. Give the probability p_+ (p_-) that a spin is in an energy state $\varepsilon_+ = -\mu B$ ($\varepsilon_- = \mu B$) and the corresponding average number of atoms $\bar{N}_+$ and $\bar{N}_-$. Deduce the expression for the average energy E .
2. Calculate the partition function for the N atoms $Z_N(\beta, B)$ using arguments at fixed N_+ . Find the previously established expression for E .
3. Define $m = p_+ - p_-$ and show that

$$\beta\mu B = \tanh^{-1} m = \frac{1}{2} \ln \frac{1+m}{1-m}$$

Verify that the entropy of a spin, S/N , may be written in the form

$$S/N = -k \left[\frac{1+m}{2} \ln \frac{1+m}{2} + \frac{1-m}{2} \ln \frac{1-m}{2} \right] \quad (3.152)$$

and show that this expression could have been obtained directly from (2.51).

4. We now assume that the energy of the spin system is fixed. Starting with (3.152), plot S as a function of $E/N\mu B$ and show that there is a zone of negative temperatures. Plot the magnetization $\mathcal{M} = Nm\mu$ as a function of $\beta\mu B$. Verify that the negative temperatures are related to negative magnetizations.

Calculate explicitly the equilibrium temperature $T(E)$ reached by the spin system and verify that it can be written as

$$\beta(E) = \frac{1}{kT(E)} = -\frac{1}{\mu B} \tanh^{-1} \frac{E}{N\mu B}$$

5. We have two such crystals, one with N_1 moments μ_1 , the other with N_2 moments μ_2 , and we take $N_1\mu_1 > N_2\mu_2$. The two crystals are initially isolated and independent at equilibrium with inverse temperatures $\beta_1(E_1)$ and $\beta_2(E_2)$. We put

the crystals in thermal contact and *consider only the interactions between the two spin systems*. Plot on the same figure the curves $\beta_1(E_1)$ and $\beta_2(E_2)$. Use a graphical analysis like we did for Figure 1.5 to show that if we maintain the point of view that the heat flow takes place from the hotter to the colder system, we are led to the conclusion that negative temperatures are hotter than positive ones!

6. At equilibrium and in the limit $|\beta_1\mu_1 B| \ll 1$ and $|\beta_2\mu_2 B| \ll 1$, show that the energies E'_1 and E'_2 of the two crystals are related by

$$\frac{E'_1}{E'_2} = \frac{N_1}{N_2} \left(\frac{\mu_1}{\mu_2} \right)^2$$

Let Q_1 be the amount of energy absorbed by crystal 1 in the form of heat. Show that

$$Q_1 = \frac{N_2(\mu_2 B)^2}{1 + \frac{N_2}{N_1} \left(\frac{\mu_2}{\mu_1} \right)^2} (\beta_1 - \beta_2) \quad (3.153)$$

Verify that this expression gives the temperature order established previously.

3.8.3 Diatomic molecules

We reconsider the problem of diatomic molecules (Section 3.3.2) by taking into account nuclear degrees of freedom due to nuclear spins. Equation (3.64) for the partition function should be modified to include the contributions of the spins embodied in ζ_{nuc}

$$Z_N(\beta) = \frac{1}{N!} (\zeta(\beta) \zeta_{\text{rot}} \zeta_{\text{vib}} \zeta_{\text{nuc}})^N \quad (3.154)$$

The spins of the two nuclei forming the molecule are written $\vec{F}$ and $\vec{F}'$. These spins interact with each other and with the electrons giving energy levels that depend on the total spin $\vec{F}_{\text{tot}} = \vec{F} + \vec{F}'$. However, the energies involved in these levels are of the order of 10^{-6} eV.

1. Show that for temperatures of the order of a kelvin or higher, the entropy of the nuclear spins is $kN \ln[(2F+1)(2F'+1)]$, where F and F' are integers or half integers. Deduce from this the expression for ζ_{nuc} when $T \geq 1$ K.

2. We now consider exclusively the hydrogen molecule where the nuclei are either protons H (spin-1/2) or deuterons D (spin-1). We recall that the D mass is twice the H mass with a precision of 10^{-3} . We will consider only temperatures ranging from about 20 K to the ambient temperature (300 K). For the H_2 molecule, the rotation temperature T_{rot} ($kT_{\text{rot}} = \hbar^2/2I$) is about 85.3 K whereas the vibration

temperature T_{vib} ($kT_{\text{vib}} = \hbar\omega_{\text{vib}}$) is about 6125 K. Show that in the temperature range of interest, we may take $\zeta_{\text{vib}} = \text{const}$. Calculate T_{rot} and T_{vib} for the HD and D_2 molecules.

3. The fact that the H_2 molecule has two identical nuclei leads to restrictions on the allowed values of J , the rotational angular momentum, due to the symmetry properties of the total wave function under exchange of the nuclei. The total nuclear spin can take the values $F_{\text{tot}} = 1$ and $F_{\text{tot}} = 0$. Show that $F_{\text{tot}} = 1$ leads to odd values for J (ortho-hydrogen) while $F_{\text{tot}} = 0$ leads to even values for J (para-hydrogen). Show that, in the temperature range of interest, the partition function may now be written as

$$Z_N(\beta) = \frac{1}{N!} (\zeta(\beta) \zeta_{\text{rot,nucl}})^N \quad (3.155)$$

and give the expression for $\zeta_{\text{rot,nucl}}$ with explicit calculation.

4. Show that for $T \ll T_{\text{rot}}$ we can write

$$\zeta_{\text{rot,nucl}} \simeq 1 + A \exp(-B/T) \quad (3.156)$$

and determine A and B . Calculate $\zeta_{\text{rot,nucl}}$ for $T \gg T_{\text{rot}}$. Show that the resulting expression is half of what we would have obtained had we ignored the indistinguishability of the two nuclei. Can you interpret this result? Give the expressions for the specific heat for $T \ll T_{\text{rot}}$ and $T \gg T_{\text{rot}}$.

5. Let $r(T)$ be the ratio of the para-hydrogen population to the total number of molecules

$$r(T) = \frac{N_{\text{para}}}{N_{\text{para}} + N_{\text{ortho}}} \quad (3.157)$$

What are the limiting values of $r(T)$ for $T \ll T_{\text{rot}}$ and $T \gg T_{\text{rot}}$? Calculate numerically $r(T)$ for $T = T_{\text{rot}}$. Sketch a representative curve for $r(T)$.

3.8.4 Models of a boundary surface

We wish to model an interface, for example the interface between the liquid and gaseous phases. For simplicity, we consider a two-dimensional model where *the interface is a line* and not a surface (Figure 3.19).

A Continuum model

1. The interface is a continuous line given by the differentiable single valued function $y(x)$ (Figure 3.19), and we assume that $(dy/dx)^2 \ll 1$. The energy of the interface is proportional to its length $\mathcal{L}$: $H = \alpha(\mathcal{L} - L)$, where α is a positive

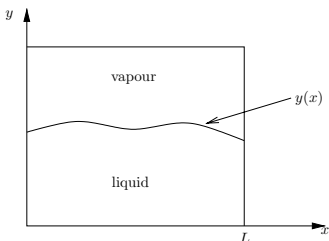


Figure 3.19 Continuum model of a one-dimensional interface.

constant, L the length of the container and αL is the energy of the shortest interface which is taken as the reference energy. Express $\mathcal{L}$ in terms of L and $(dy/dx)^2$.

2. We take the boundary conditions: $y(0) = y(L) = 0$. We can then write $y(x)$ in terms of its Fourier components

$$y(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{\pi n x}{L}\right) \quad (3.158)$$

Show that H can be put in the form

$$H = \frac{\alpha \pi^2}{4L} \sum_{n=1}^{\infty} n^2 A_n^2 \quad (3.159)$$

3. Find, up to a multiplicative constant, the probability $P(A_n)$ to observe the mode n and the joint probability $P(A_n, A_m)$ to observe two modes. Use the result to obtain $\langle A_n \rangle$ and $\langle A_n A_m \rangle$.

4. Let $\Delta y = y(x) - y(x')$ be the interface height difference at the two points x and x' . When the two points x and x' are well separated, the average $\langle (\Delta y)^2 \rangle^{1/2}$ may be considered as the width of the interface. Calculate $\langle y(x)y(x') \rangle$, say for $x' \geq x$, and use that to obtain $\langle (\Delta y)^2 \rangle$. The following identity may be useful

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \cos nx = \frac{\pi^2}{6} - \frac{\pi|x|}{2} + \frac{x^2}{4} \quad 0 \leq x \leq 2\pi \quad (3.160)$$

B Discrete Ising model

We now use a two-dimensional Ising model where the interface is the line separating a region of positive magnetization (Ising spins $+$) from a region of negative

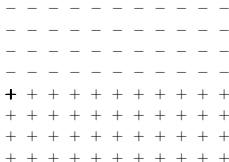


Figure 3.20 The ground state of the model.

magnetization (Ising spins $-$) (Figure 3.20). The spins are arranged on an $N \times N$ square lattice with unit spacing and $N \gg 1$. The system Hamiltonian is

$$H = -J \sum_{\langle i,j \rangle} S_i S_j \quad S_i, S_j = \pm 1$$

where J is a positive constant and the sum $\sum_{\langle i,j \rangle}$ runs over near neighbours.

1. What does the quantity $\sum_{p=1}^N (1 + |y_p|)$ represent, where y_p is the height of the step at point x_p (for example, in Figure 3.21, $y_1 = 2$, $y_2 = 1$, $y_3 = 0$, $y_4 = -2$, etc.)? Show that the energy due to the presence of an interface (i.e. the energy difference between the ground state and a generic situation) is given by

$$H = 2JN + 2J \sum_{p=1}^N |y_p|$$

2. We now assume that the left extremity of the interface is fixed while the right one is free to move. The heights y_p are assumed to be integer random variables ranging from $-\infty$ to $+\infty$. Show that the partition function for the interface is given by

$$Z = \zeta^N$$

where ζ is the partition function for a single step and is given by

$$\zeta = e^{-2\beta J} \coth(\beta J)$$

Calculate the surface tension, i.e. the free energy per unit length $f = F/N$, and the internal energy per unit length $\varepsilon = E/N$. Calculate the limits of f and ε for $\beta J \ll 1$ and $\beta J \gg 1$ and comment on the results.

3.8.5 Debye–Hückel approximation

An electrolyte is made of an equal number of positive and negative ions (charge $\pm q$) thus assuring global electric neutrality of the solution. On average over the whole volume of the solution, the densities of the two types of ions are equal (n), which means that the total ion density is $2n$. However, there are local fluctuations in the densities. To take them into account, we choose one particular ion as the origin of a reference frame. This ion will attract ions of opposite charge in its neighbourhood while repelling those with the same charge. Thus, the ion at the origin creates around it an ‘ion cloud’, which is spherically symmetric on average but inside of which the charge distribution is not uniform. We introduce the local charge densities $n^+(\vec{r})$ of ions of the same charge as the ion at the origin, and $n^-(\vec{r})$ of ions of opposite charge. The vector $\vec{r}$ gives the position with respect to the ion at the origin. We assume that deviations from the uniform situation are small, i.e. $|n^\pm(\vec{r}) - n| \ll n$, and to fix ideas we consider the ion at the origin to have the charge $+q$.

1. Let $\Phi(\vec{r})$ be the average electric potential inside the ionic cloud. The assumption of small deviations translates into the following condition: $|q\Phi(\vec{r})|/kT \ll 1$. Justify qualitatively this approximation and show that it allows us to write

$$n^\pm(\vec{r}) \simeq n \left(1 \mp \frac{q}{kT} \Phi(\vec{r}) \right)$$

2. The charge densities are also related by the Poisson equation, which we can write in the following two forms

$$\begin{aligned} -\nabla^2 \left(\Phi(\vec{r}) - \frac{q}{4\pi\epsilon_0 r} \right) &= \frac{q}{\epsilon_0} (n^+(\vec{r}) - n^-(\vec{r})) \\ -\nabla^2 \Phi(\vec{r}) &= \frac{q}{\epsilon_0} (n^+(\vec{r}) - n^-(\vec{r})) + \frac{q}{\epsilon_0} \delta(\vec{r}) \end{aligned}$$

Comment on this equation and use the results of the previous question to rewrite it as a closed equation for Φ . Let $\tilde{\Phi}(\vec{p})$ be the Fourier transform of $\Phi(\vec{r})$

$$\begin{aligned} \tilde{\Phi}(\vec{p}) &= \int d^3r e^{i\vec{p}\cdot\vec{r}} \Phi(\vec{r}) \\ \Phi(\vec{r}) &= \frac{1}{(2\pi)^3} \int d^3p e^{-i\vec{p}\cdot\vec{r}} \tilde{\Phi}(\vec{p}) \end{aligned}$$

Calculate $\tilde{\Phi}(\vec{p})$ and use the result to obtain $\Phi(\vec{r})$

$$\Phi(\vec{r}) = \frac{q}{4\pi\epsilon_0 r} e^{-r/b} \quad b = \sqrt{\frac{\epsilon_0 kT}{2nq^2}}$$

Note: It is easy to calculate the inverse transform of $\tilde{\Phi}(\vec{p})$ using the method of residues. However, we give the following integral

$$\frac{1}{(2\pi)^3} \int d^3p \frac{e^{-i\vec{p}\cdot\vec{r}}}{\vec{p}^2 + 1} = \frac{1}{4\pi r} e^{-r} \quad (3.161)$$

3. What is the total charge inside a sphere of radius R centred on the ion at the origin? It may be useful to note that $r e^{-r/b} = b^2(\partial/\partial b)e^{-r/b}$. What happens when $R \gg b$? Give a physical interpretation of your results, in particular the parameter b , called the *Debye length*.

4. Let $n_2^+(\vec{r}, \vec{r}')$ be the pair density for ions of the same charge and $n_2^-(\vec{r}, \vec{r}')$ be that for opposite charges (cf. Section 3.4.1). Justify the following expression for the electrostatic potential energy

$$E_{\text{pot}} = \frac{q^2}{4\pi\epsilon_0} \int d^3r d^3r' \frac{1}{|\vec{r} - \vec{r}'|} (n_2^+(|\vec{r} - \vec{r}'|) - n_2^-(|\vec{r} - \vec{r}'|)) \quad (3.162)$$

Show that

$$n_2^\pm(\vec{r}, \vec{r}') = nn^\pm(\vec{r}' - \vec{r})$$

and use this result to calculate explicitly the potential energy

$$E_{\text{pot}} = -nV \frac{q^2}{4\pi\epsilon_0 b}$$

5. Let $\bar{r} \sim n^{-1/3}$ be the average distance between the ions. Verify that the condition $|q\Phi(\bar{r})|/kT \ll 1$ is equivalent to $\bar{r}/b \ll 1$. What is the physical interpretation of this relation?

3.8.6 Thin metallic film

We use a simple model to study the formation of thin metallic films deposited on a flat substrate. We can, for example, make thin gold films deposited on silicon by using an oven whose walls are made of silicon and into which we introduce a gas of gold whose temperature and pressure may be varied.

We use the following model to describe the thin film (Figure 3.22):

- (i) the substrate is made of A sites ($A \gg 1$) on which the metallic atoms can be deposited,
- (ii) the deposited atoms are spinless and their position completely determines their state,
- (iii) on a given site, the atoms can pile up on top of each other without limit; each deposited atom has an energy $-\epsilon_0$ ($\epsilon_0 > 0$),
- (iv) we ignore interactions between atoms deposited on neighbouring sites,
- (v) each deposited atom contributes a height a to the thickness h of the film below it.

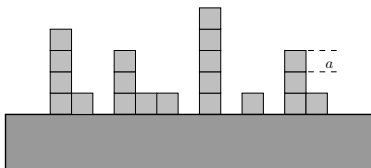


Figure 3.22 Deposition of a thin metallic film.

The gas of metal atoms (mass m and zero spin) is assumed to be ideal and classical and provides the system with a reservoir of atoms at chemical potential μ and temperature T .

1. Express the chemical potential μ of the gas in terms of its temperature T and pressure P .
2. Let $\mathcal{Q}$ be the grand canonical partition function for all of the sites and ξ that of a single site. Justify the following relation

$$\mathcal{Q} = \xi^A \quad (3.163)$$

and calculate ξ . Show that the chemical potential cannot exceed a limiting value to be determined and give the expression for $P_0(T)$, the pressure at this limit value.

3. Calculate $\overline{N}_0$, the average number of empty sites and $\langle h \rangle$, the average thickness of the film. We define the relative average deviation of the height of the film by

$$\frac{\Delta h}{\langle h \rangle} = \frac{(\langle h^2 \rangle - \langle h \rangle^2)^{1/2}}{\langle h \rangle}$$

Calculate this quantity.

4. Express the average height $\langle h \rangle$ in terms of P and P_0 and plot its behaviour as a function of P for two temperatures T_1 and T_2 ($T_1 < T_2$).
5. Express $\overline{N}_0$ as a function of P and P_0 and plot the behaviour as a function of P at a given temperature. Do the same for the quantity $\Delta h / \langle h \rangle$.
6. It can be shown that the metal film deposited on an insulating substrate conducts electricity when the number of empty sites is less than a known value x_0 called the percolation threshold. What is the minimum average thickness of the film for it to be conducting?

3.8.7 Beyond the ideal gas: first term of virial expansion

We consider a *classical* mono-atomic gas whose partition function $Z_N(V, T)$ may be written in the form

$$Z_N(V, T) = Z_K(V, T) Z_U(V, T) \quad (3.164)$$

where $Z_K(V, T)$ is the ideal gas partition function and

$$Z_U(V, T) = \frac{1}{V^N} \int d^3r_1 \dots d^3r_N \exp \left[-\frac{\beta}{2} \sum_{i \neq j} U(|\vec{r}_i - \vec{r}_j|) \right] \quad (3.165)$$

The potential energy $U(|\vec{r}_i - \vec{r}_j|)$ of two molecules at $\vec{r}_i$ and $\vec{r}_j$ is a function only of the distance between them $|\vec{r}_i - \vec{r}_j|$. We shall study below the corrections to the ideal gas equation of state by obtaining a power series in the density $n = 1/v = N/V$. In this problem we stop at first order in $1/v$ but it is possible to derive a systematic expansion, called the virial expansion.

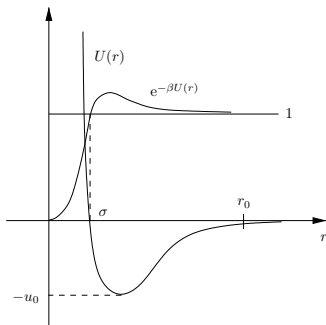
1. Show that the equation of state giving the pressure P as a function of v is

$$\frac{Pv}{kT} = 1 + \frac{V}{N} \frac{\partial \ln Z_U}{\partial V} = 1 + \frac{v}{N} \frac{\partial \ln Z_U}{\partial v} \quad (3.166)$$

2. We consider Z_U under the assumption that the gas is very dilute. It is convenient to write Z_U in the form

$$Z_U(V, T) = 1 + \frac{1}{V^N} \int d^3r_1 \dots d^3r_N \left(\exp \left[-\frac{\beta}{2} \sum_{i \neq j} U(|\vec{r}_i - \vec{r}_j|) \right] - 1 \right) \quad (3.167)$$

From the form of the intermolecular potential (see Figure 3.23), it is clear that $\exp[-\beta U(r)]$ is different from 1 only for distances less than the range of the potential r_0 . Sketch qualitatively $\exp[-\beta U(r)]$ for $T_1 \ll T$ and $T_2 \gg T$ where T is the temperature we used to obtain the curves in Figure 3.23. Let us consider a gas molecule and ignore interactions. What is the average number of molecules in the sphere of radius r_0 centered on this molecule? If the gas is very dilute, this number is $\ll 1$. What is the probability of finding another molecule at a distance $< r_0$ of the molecule at the centre? Under these conditions, what is the average number of pairs of molecules with a separation between the constituent molecules $< r_0$? What is the condition for this average number to be $\ll 1$? Show that if this

Figure 3.23 The potential $U(r)$ and $\exp[-\beta U(r)]$.

condition is satisfied we have

$$Z_U(V, T) \simeq 1 + \frac{N^2}{2V} \int d^3r \left(e^{-\beta U(r)} - 1 \right) \quad (3.168)$$

$$= 1 - \frac{N^2}{V} B(T) \quad (3.169)$$

where $B(T)$, the second virial coefficient, is defined by the second equation. Show that the free energy is extensive and verify that, with the above approximations for Z_U , the equation of state becomes

$$\frac{Pv}{kT} = 1 + \frac{B(T)}{v} \quad (3.170)$$

3. Show that in the low density limit, the van der Waals equation (1.66)

$$\left(P + \frac{a}{v^2} \right) (v - b) = kT \quad (3.171)$$

reduces to an equation of the above type. Calculate $B(T)$ as a function of a , b and T . Can you give a physical interpretation of the result?

4. Another demonstration of (3.170). We start with Equation (3.87) for the exact expression for the pressure

$$P = nkT - \frac{1}{6}n^2 \int d^3r r U'(r) g(r) \quad (3.172)$$

Table 3.1 Values of the parameter $B(T)$ for nitrogen at various temperature values.

T (K)	100	200	300	400	500
$B(T)$ ($\text{\AA}^3$)	-247	-58.4	-7.5	15.3	28.1

where g is the pair correlation function (3.77). Show that for a dilute gas we have

$$g(r) \simeq \exp[-\beta U(r)]$$

and re-obtain Equation (3.170) for the pressure. Why is the above equation for $g(r)$ not valid for a dense fluid?

5. What is the value of $B(T)$

(i) for a gas of hard spheres

$$U(r) = \begin{cases} +\infty & \text{if } r \leq \sigma \\ 0 & \text{if } r > \sigma \end{cases}$$

(ii) for an inter-molecular potential given by

$$U(r) = \begin{cases} +\infty & \text{if } r \leq \sigma \\ -u_0 & \text{if } \sigma \leq r \leq r_0 \\ 0 & \text{if } r > r_0 \end{cases}$$

How does the sign of $B(T)$ depend on the temperature? Can you give an interpretation for the change in the sign of $B(T)$? Hint: Examine the cases $T \rightarrow 0$ and $T \rightarrow \infty$.

(iii) for a Lennard-Jones potential

$$U(r) = 4u_0 \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]$$

Show that $B' = B/\sigma^3$ is a universal function of the dimensionless parameter $\theta' = kT/u_0$, independent of the gas. Experiments show that hydrogen and helium do not fall on this universal curve at low temperature. Why?

6. The coefficient $B(T)$ has been measured for nitrogen as a function of T (although nitrogen is diatomic, it can be shown that the above arguments still hold). Calculate numerically $B(T)$ for the Lennard-Jones potential with the values $u_0/k = 95$ K and $\sigma = 3.74 \times 10^{-10}$ m and compare with the values in Table 3.1.

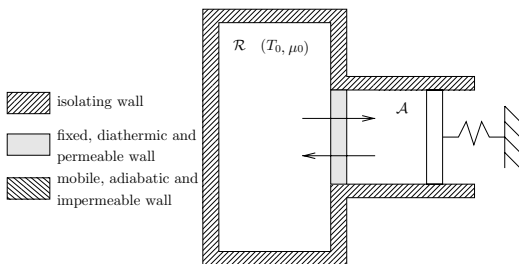


Figure 3.24 Schematic representation of the system under consideration.

3.8.8 Theory of nucleation

We consider a system $\mathcal{A}$ that can exchange particles and heat with a reservoir $\mathcal{R}$ at temperature T_0 and chemical potential μ_0 (Figure 3.24). The transformations of the reservoir will always be considered quasi-static. The system can, in addition, exchange energy in the form of work with the external world (represented by the spring in Figure 3.24).

1. Consider the case where the total system ' $\mathcal{A} + \mathcal{R}$ ' is isolated from the external world, i.e. it does not exchange work with it. Show that the change in the quantity $(E - T_0S - \mu_0N)$ is negative when the system $\mathcal{A}$ is taken from an initial equilibrium state (i) to a final equilibrium state (f) :

$$\Delta(E - T_0S - \mu_0N) \leq 0$$

In this equation, E , S and N are respectively the internal energy, the entropy and the number of particles of the system $\mathcal{A}$ (we use the same notations as in Sections 3.5.3 and 3.5.4).

2. Suppose that during a transformation $(i) \rightarrow (f)$ the system $\mathcal{A}$ receives an amount of work W from the external world. Show that

$$W \geq W_{\min} = \Delta(E - T_0S - \mu_0N) \quad (3.173)$$

where $W_{\min}$ is the work received during a reversible transformation $(i) \rightarrow (f)$.

3. Figure 3.25 shows, as a function of its total internal energy E_{tot} , the entropy S_{tot} of the total system at equilibrium. Justify qualitatively the form of the curve $S_{\text{tot}}(E_{\text{tot}})$.

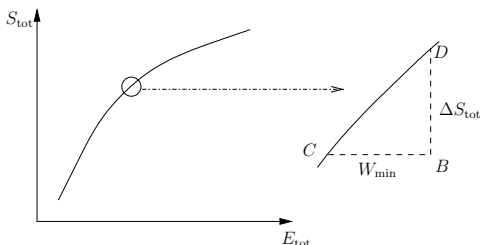


Figure 3.25 Entropy versus energy for the total system.

4. Starting in a situation of global equilibrium (D), a fluctuation in the total entropy can take the system to a non-equilibrium state (B). The point B may also be reached via an isentropic transformation by providing the system $\mathcal{A}$ (now isolated from $\mathcal{R}$) an amount of work $W_{\min}$ along the path $C \rightarrow B$. Assume that $W_{\min}$ is small compared to $E_{\text{tot}}(D)$. By using the entropy–energy curve, establish a relation between $W_{\min}$ and ΔS_{tot} . Show that the probability $\mathcal{P}$ of such a fluctuation is

$$\mathcal{P} \propto \exp[-W_{\min}/kT_0]$$

It will be useful to recall that for an isolated system, the entropy is given by k times the logarithm of the number of accessible configurations.

5. We now consider the formation of a droplet in a supersaturated vapour in metastable equilibrium at temperature T . At this temperature, the pressure at the liquid–vapour transition is $\bar{P}$, the vapour pressure is P_2 and the pressure at the interior of the liquid droplet is P_1 (Figure 3.26). Show that the formation of a droplet of volume V_1 is accompanied by a change in the grand potential given by

$$\Delta\mathcal{J} = V_1(P_2 - P_1)$$

The above expression corresponds to a situation where the gas–liquid interface is plane. To include the effect of the shape of the droplet, one should include a surface energy term, σA , where σ is the surface tension

$$\Delta\mathcal{J} = V_1(P_2 - P_1) + \sigma A$$

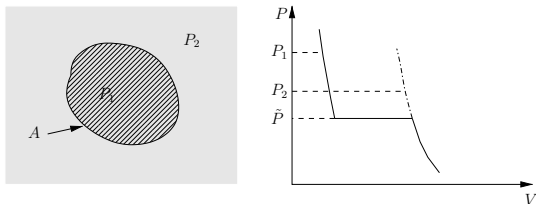


Figure 3.26 Phase diagram.

Argue that $\Delta\mathcal{J}$ corresponds to the minimum work to be furnished to the vapour to form the droplet

$$W_{\min} = \Delta\mathcal{J}$$

6. Calculate $W_{\min}$ for a spherical droplet of radius R . Plot $W_{\min}$ as a function of R and verify that it passes through a maximum for $R = R^*$ to be determined. What happens if the radius of the droplet is larger than the critical radius? Smaller?
7. Calculate the probability for the formation of a droplet of critical radius.
8. We take $\delta P_1 = P_1 - \tilde{P}$ and $\delta P_2 = P_2 - \tilde{P}$ ($|\delta P_1|, |\delta P_2| \ll \tilde{P}$). By using the Gibbs–Duhem relation, show that

$$v_1 \delta P_1 = v_2 \delta P_2$$

and that

$$R^* = \frac{2\sigma v_1}{\mu_2(P_2, T) - \mu_1(P_2, T)} \quad (3.174)$$

where v_1 and v_2 are the specific volumes of the liquid and gas phases respectively.

3.9 Further reading

The results presented in this chapter are classic and appear in all books on statistical mechanics. For additional reading we recommend Reif [109] (Chapter 7). The Ising model in one dimension is treated in detail by Le Bellac [72] (Chapter 1); the solution in two dimensions (difficult!) is described in Onsager's article [97]. See also Huang [57] (Chapter 17) and Landau and Lifshitz [70] (Section 151); the historical aspects of the model are discussed by Brush [22]. For a demonstration of the Lee–Yang theorem see Huang [57] (Appendix C) or Ma [85] (Chapter 9). A

heuristic demonstration of the existence of a thermodynamic limit in the presence of Coulomb interactions is given by Lévy-Leblond and Balibar [80] (Chapter 6); the rigorous proof is due to Dyson and Lenard [36]. The theory of classical liquids is treated in Chandler [28] (Chapter 7), McQuarrie [88] (Chapter 13), Goodstein [49] (Chapter 4) and, at a more advanced level, Hansen and McDonald [54]. The use of the chemical potential in phase transitions is described by Schroeder [114] (Chapter 5), or Chaikin and Lubensky [26] (Section 4.4). Its use in chemical reactions is discussed in Reif [109] (Chapter 8), Baierlein [4] (Chapter 11) and Brenig [19] (Chapter 20). Equivalence between the canonical and grand canonical ensembles is treated in Huang [57] (Chapter 8), which also gives a short account of second quantization in Appendix A.

Additional information on the problems in this chapter can be found in the following references. The original experiment on negative temperatures was done by Purcell and Pound [105]. The Debye–Hückel approximation is discussed in Landau and Lifshitz [70] (Section 78). Further information on the virial expansion is found in Landau and Lifshitz [70] (Section 74) and in McQuarrie [88] (Chapter 12). For the theory of nucleation, consult Landau and Lifshitz [70] (Section 162).